

A Survey on Deep Reinforcement Learning Applications in Autonomous Systems: Applications, Open Challenges, and Future Directions

Shruti Govinda¹, Bouziane Brik², *Senior Member, IEEE*, and Saad Harous³, *Senior Member, IEEE*

Abstract—Deep Reinforcement Learning (DRL) has become a fundamental element in advancing Autonomous Systems, significantly transforming fields like autonomous vehicles, robotics, and drones. This survey paper provides a comprehensive overview of the role of DRL in autonomous systems, focusing on recent advancements, applications, and challenges. Through a synthesis of existing literature and case studies, the paper elucidates key principles, methodologies, and implications of integrating DRL into autonomous systems. The systematic examination of selected papers reveals recurring patterns, emerging trends, and identifies gaps and opportunities for further research. By exploring the applications of DRL across different autonomous systems, commonalities, distinctions, and prevalent challenges are discussed, laying the groundwork for future advancements and practical implementations in this rapidly evolving field.

Index Terms—Deep reinforcement learning (DRL), autonomous systems, autonomous cars, autonomous robotics, autonomous drones.

I. INTRODUCTION

A. Context and Motivation

IN RECENT years, autonomous systems have made remarkable progress, revolutionizing industries such as robotics, transportation, manufacturing, and healthcare. These systems, engineered to operate and make decisions autonomously, offer vast potential for enhancing efficiency, safety, and productivity across diverse domains [1]. Central to the development of autonomous systems is the integration of Deep Reinforcement Learning (DRL), a potent framework that merges reinforcement learning techniques with deep neural networks [2]. DRL empowers autonomous agents to learn optimal decision-making strategies through interaction with their surroundings, facilitating adaptation to dynamic and complex environments.

Received 9 September 2024; revised 21 January 2025; accepted 11 March 2025. Date of publication 8 May 2025; date of current version 1 July 2025. This work was supported by the French Region Project, Accueil de Nouvelle Équipe de Recherche (ANER), 5G-VIA. The Associate Editor for this article was Z. Gu. (*Corresponding author: Shruti Govinda.*)

Shruti Govinda and Saad Harous are with the Computer Science Department, College of Computing and Informatics, University of Sharjah, Sharjah, United Arab Emirates (e-mail: u23106215@sharjah.ac.ae; harous@sharjah.ac.ae).

Bouziane Brik is with the Computer Science Department, College of Computing and Informatics, University of Sharjah, Sharjah, United Arab Emirates, and also with the Université Bourgogne Europe, DRIVE UR 1859, 58000 Nevers, France (e-mail: bbrik@sharjah.ac.ae).

Digital Object Identifier 10.1109/TITS.2025.3560379

Its significance in autonomous systems cannot be overstated, as it enables adaptive decision-making, continual learning, and resilience, ensuring effectiveness in real-world contexts [3].

The motivation to explore DRL further stems from its profound implications across a spectrum of applications, ranging from autonomous cars to robotics and autonomous drones, as well as Advanced Driver Assistance Systems (ADAS). Recent years have witnessed a surge of interest and innovation in the application of autonomous systems across various sectors, driven by groundbreaking advancements in DRL [4]. This powerful framework has emerged as a cornerstone in the development of autonomous technologies, offering the capability to make optimal decisions without direct human intervention.

Key driving forces behind the growing interest in DRL include its ability to facilitate adaptive decision-making and learning from experience in complex and dynamic environments [5]. Autonomous agents trained with DRL can learn optimal strategies, enabling them to navigate intricate scenarios with ease and efficiency. This adaptability is particularly crucial in domains like autonomous cars and ADAS, where real-world conditions are inherently unpredictable and constantly changing.

Moreover, DRL empowers autonomous systems, including ADAS, to learn from experience, enhancing their performance and resilience over time [6]. By continually refining their decision-making processes based on feedback from the environment, these systems can evolve to tackle novel challenges and scenarios, ensuring robustness and reliability in operation. The versatility of DRL extends beyond individual applications to encompass a wide array of autonomous systems, including robotics and autonomous drones. Recent research has showcased the efficacy of DRL in optimizing manufacturing processes, enhancing the agility and precision of robotic manipulation tasks, and enabling autonomous drones to navigate complex environments with unparalleled efficiency [7], [8].

This growing interest in DRL is not confined to academic research alone; it is increasingly gaining traction in industrial and commercial settings as well. Companies across various sectors recognize the transformative potential of DRL in enhancing the capabilities of autonomous systems, leading to

substantial investments and collaborations aimed at harnessing this technology for practical applications [9].

B. Review of Existing Related Surveys

The application of Deep Reinforcement Learning (DRL) has been investigated in numerous surveys for various domains, where each domain brings unique challenges and opportunities. For instance, in [10], the paper provides a comprehensive review of DRL in mobile robot navigation. It compares DRL with traditional navigation technology and Simultaneous Localization and Mapping (SLAM), highlighting the advantages of DRL. The document systematically elaborates on the background and classic algorithm models of standard reinforcement learning and DRL. It discusses the application of DRL in different scenarios, such as indoor and crowd navigation, addressing challenges and presenting innovative solutions. The paper emphasizes the potential of DRL in improving current research results and introducing DRL into more practical applications. Furthermore, [11] gives a study on navigation for agents utilizing deep reinforcement learning discusses the evolution of reinforcement learning, deep learning, and their integration into DRL. It classifies DRL visual navigation approaches into five categories: hierarchical, multi-task, memory-inference, and vision-language. The authors tackle issues like data inefficiency and limited generalization in settings and suggest avenues for further research such as policy hierarchy, meta-learning, memory utilization, and multi-modal fusion to improve visual navigation capabilities. Similarly, a comprehensive review [12] of DRL in manipulation delves into the core principles of reinforcement learning, cutting-edge DRL algorithms, and the obstacles and solutions in deploying these algorithms in robotics. This document explores techniques for reward design, network structures, and notable advancements in tasks like grasping and object handling while highlighting the significance of efficient sample usage, transfer learning opportunities, real-time control mechanisms, and safe exploration strategies. It also discusses integration with other educational frameworks for prospective investigations.

Transitioning to autonomous driving, another study [13] provides a comprehensive overview of the application of reinforcement learning (RL) in autonomous driving systems. It discusses the challenges, opportunities, and future research directions in this emerging field. The authors present a detailed theoretical reinforcement learning framework and conduct a literature survey on applying RL for autonomous driving tasks. They address challenges like bridging the simulation-reality gap and sample efficiency in RL. The paper highlights the need for validation of RL-based systems, designing good reward functions, and incorporating safety into decision-making RL systems for autonomous agents. It also discusses open-source frameworks and packages for state-of-the-art RL/DRL algorithms and evaluation. The importance of multi-agent reinforcement learning approaches and the implementation of RL algorithms is emphasized. Furthering the role of DRL in autonomous vehicles, a paper [14] surveys DRL applications in path planning and control, discussing how DRL methodologies and algorithms integrate into trajectory planning and dynamic control. It reviews the three-tier control

framework of autonomous vehicles—perception, planning, and control—and highlights the limitations of traditional control methodologies, suggesting DRL as a solution to complex control challenges in autonomous driving. This paper further discusses the use of algorithms such as SAC, PPO, and DDPG in various driving subfields like intersection management, urban navigation, and highway motoring, emphasizing the importance of state and action design and crafting reward functions. The integration of DRL techniques becomes crucial in advancing autonomous vehicle control, as demonstrated in a study [15], which inspects the performance regarding effective interaction with the environment of DQN and DDPG algorithms. This work clarifies insights into optimal policies, showing DDPG's better performance in maintaining lane and velocity compared to DQN, which shows better overall success. These results significantly contribute to DRL's practical application in autonomous navigation, elucidating ways for further development in this field. Another comprehensive survey [16] on DRL in the context of autonomous vehicle motion planning discusses challenges and progress using DRL for different driving scenarios: car-following, lane-keeping, merging, and navigating in dense traffic. It highlights the importance of multi-agent reinforcement learning (MARL) for interactions with other vehicles and examines different approaches to MARL, such as parameter sharing, centralized control, and coordination graphs. This survey also includes the theory of safe reinforcement learning, Sim2Real knowledge transfer, and the use of various simulators, including SUMO and TORCS. Furthermore, the paper [17] surveys deep learning applications in autonomous vehicle control, categorizing them into lateral (steering), longitudinal (acceleration and braking), and simultaneous lateral and longitudinal control. It discusses the strengths and limitations of various methods, highlighting challenges such as computation, architecture selection, goal specification, generalization, verification, validation, and safety. The trend towards deeper models with more training data and the introduction of temporal cues, despite training instability, is noted. Most models are trained and evaluated in simple environments, often focusing on single tasks like lane keeping without addressing complex decision-making like lane changes or turns. The authors provide an overview of relevant techniques, learning strategies, datasets, and tools, emphasizing the need for extensive real-world testing before deployment. Specifically, the paper discusses Inverse Reinforcement Learning (IRL) for defining optimal reward functions from expert demonstrations, and the use of Deep Q Networks (DQN) and Deep Deterministic Policy Gradient (DDPG) in vehicle control. The survey concludes that while deep learning shows promise, significant improvements in controller performance and extensive real-world testing are essential for validation and safety assurance.

Regarding autonomous drones, the paper [18] examines the application of DRL in multi-drone systems, focusing on scalability and cooperation without centralized control during execution. The authors conducted a comprehensive search for relevant literature, applying stringent filters to select studies that met their criteria for cooperative and scalable approaches. They categorized the selected papers into five

drone application classes: coverage, adversarial search and game, computational offloading, communication, and target-driven navigation. The review highlights the importance of DRL in addressing the challenges of multi-drone systems and provides a critical analysis of the current state of research, emphasizing the need for further exploration in this field to enhance the integration of drones into society, especially in applications that could impact privacy and public perception. Similarly, the paper [15] provides a comprehensive review of DRL techniques in drones. It addresses challenges faced by drones in path planning, navigation, and control, discussing the use of reinforcement learning algorithms to navigate through unknown environments. The document highlights various DRL methods designed to ensure stable and smooth drone navigation and emphasizes the need for further research to address challenges that restrain their deployment in real-life scenarios. Additionally, the paper delves into the use of DRL in important applications like wildfire monitoring, rescue missions, and dangerous environment exploration, discussing the potential of DRL in enhancing drone control systems and the ongoing development of new DRL methods and strategies.

Lastly, the paper [74] provides an in-depth analysis of vehicle detection methods within Advanced Driver Assistance Systems (ADAS) and autonomous vehicles. It discusses the evolution of detection techniques, from traditional feature-based and model-based methods to the advent of deep learning approaches, including deep reinforcement learning (DRL). The paper highlights the challenges of vehicle detection in various environments and the importance of efficient algorithms for generating candidate regions and robust classifiers. The authors emphasize the superiority of object proposal methods (OPMs) over the traditional sliding window approach, particularly Selective Search, which has been adapted for use with aerial images. They also explore the integration of DRL-based classifiers, such as Deep Neural Networks (DNNs) and Convolutional Neural Networks (CNNs), which have shown remarkable performance in complex detection tasks due to their ability to construct hierarchical representations and handle non-linearity. The paper presents experimental results using the VEDAI database, demonstrating the effectiveness of the adapted Selective Search method for generating object proposals and the deep learning classifier framework, enhanced with DRL techniques, for vehicle detection in aerial images. The authors conclude that while deep learning-based methods, including those utilizing DRL, outperform handcrafted feature-based classifiers, there is still a performance drop in aerial imagery due to low resolution and small vehicle sizes. In summary, the paper contributes to the field of ADAS by reviewing and comparing different vehicle detection methods, showcasing the potential of deep learning and DRL techniques, and providing insights into the challenges of aerial vehicle detection.

C. Our Contributions

This survey paper makes several significant contributions to the field of Deep Reinforcement Learning (DRL) in autonomous systems.

- **Comprehensive Overview of DRL in Autonomous Systems:** The primary contribution of this work is providing a detailed overview of the role and impact of DRL across various autonomous systems, including self-driving cars, robotics, and drones. By synthesizing recent research findings and analyzing numerous case studies, this paper highlights DRL's potential to revolutionize these systems by enabling adaptive decision-making, enhancing performance through continuous learning, and ensuring resilience in dynamic environments.
- **Identification of Key Themes and Techniques:** Another major contribution is the identification and discussion of common themes, techniques, and results prevalent in DRL research. This includes an in-depth analysis of methodologies used in experimental and testing setups, offering insights into how DRL techniques are being practically implemented and evaluated in autonomous systems.
- **Insightful Analysis of Challenges and Future Directions:** The paper provides a critical analysis of the challenges and obstacles in integrating DRL into autonomous systems, such as simulation-to-reality transfer, sample efficiency, and safety concerns. Furthermore, it suggests potential areas for improvement and innovation that could enhance the practical application of DRL in autonomous systems, thus guiding future research directions.
- **Structured Comparative Analysis:** Lastly, the paper conducts a structured comparison of the novelty and advancements of the surveyed works against existing literature. This comparative analysis aims to contextualize the contributions of recent research, helping to identify gaps in current knowledge and pointing towards new avenues for exploration.

By addressing these key areas, this survey paper aims to serve as a comprehensive resource for researchers and practitioners in the field, guiding future research and practical implementations of DRL in autonomous systems.

D. Paper Outline

As illustrated in Fig. 1, the organization of this paper is as follows: Section I sets the stage by providing the context and motivation for the investigation into Deep Reinforcement Learning (DRL) applications in autonomous systems. It also includes a review of existing related surveys and outlines the primary contributions of this paper. Section II delves into the background of DRL and its foundational concepts, as well as its advancements that are pertinent to autonomous systems. In Section III, we focus on the application of DRL within autonomous systems, specifically examining its role in autonomous cars, autonomous robotics, and autonomous drones. Section IV highlights Industrial Applications and Real-World Deployments, providing examples of how DRL has been implemented across various industries such as autonomous vehicles, robotics, and drones, and discusses the challenges and successes of these implementations. Section V addresses the open challenges and future directions for DRL in autonomous systems. This section covers key topics, including scalability and generalization of DRL techniques, safety and

robustness concerns, transitioning from simulation to real-world deployment, hybrid architectural DRL methods, the integration of large language models (LLMs) and transformer-based architectures, and ethical and regulatory considerations. Finally, the paper concludes in Section VI, summarizing the insights and findings from the reviewed literature. For quick reference, a list of acronyms used throughout the paper is provided in alphabetical order in Table II. Throughout this paper, the terms ‘Deep Reinforcement Learning’ and ‘DRL’ are used interchangeably.

E. Survey Methodology

This survey follows a systematic literature review (SLR) approach, querying major databases such as IEEE Xplore, ACM Digital Library, Google Scholar, ProQuest, EBSCO, ScienceDirect, and MDPI for relevant publications. Keywords used include “Deep Reinforcement Learning-DRL,” “autonomous systems,” “autonomous cars,” “robotics,” “drones,” “drone,” and “ADAS”. The selection criteria prioritized relevance, citation impact, and publication date, with a focus on papers published from 2018 to 2024 to cover recent advancements. The classification scheme in this survey is structured to systematically organize the reviewed papers based on their application domains and specific challenges tackled by DRL techniques. This scheme serves three primary purposes:

- *Domain-Specific Relevance:* Each autonomous system (e.g., cars, robotics, drones, ADAS) has distinct functionalities, goals, and challenges. Grouping papers by their domains ensures a targeted analysis of how DRL techniques address domain-specific needs.
- *Subtopic Categorization:* Within each domain, DRL applications are further classified into subtopics that represent common tasks or objectives. This categorization highlights the diversity of DRL applications and enables focused comparisons of methodologies and performance metrics.
- *Clarity and Accessibility:* The classification scheme enhances readability by presenting a logical flow, starting with broader domains and narrowing down to specific subtopics. This approach helps readers understand the scope of DRL applications across various autonomous systems and identify trends and research gaps.

Following this scheme, the selected papers were classified into sections and subsections based on their contributions to specific subtopics, structuring the survey for clarity. For Autonomous Cars, subtopics included Lane Following and Parking Maneuvering, Urban Navigation and Vehicle Routing, Obstacle and Collision Avoidance, and Braking System. Autonomous Robotics covered Swarm Behaviour, Collision Avoidance, and Trajectory Planning. Autonomous Drones encompassed Security and Pollution Control, Stability and Trajectory Planning, and Collision Avoidance. ADAS subtopics included Lane Departure Warning/Correction, Pedestrian Detection and Collision Avoidance, and Adaptive Cruise Control. This systematic organization of literature allows for focused comparisons within each domain,

enabling a structured comparison of methodologies, contributions, and findings. The methodology ensures a comprehensive and balanced analysis of DRL techniques applied across various autonomous systems.

II. BACKGROUND

This section provides a background on autonomous systems and Deep Reinforcement Learning (DRL), which are essential for understanding the advancements in autonomous technologies. We begin by introducing autonomous systems, highlighting their evolution and applications, to give non-expert readers a foundational understanding of this transformative technology. This is followed by an exploration of DRL, an advanced machine-learning approach crucial for enhancing the capabilities of autonomous systems in complex and dynamic environments.

A. Autonomous Systems

Autonomous systems signify a fundamental change in the execution of tasks and decision-making processes, transitioning from conventional human-centered methods to more automated and adaptable strategies. These systems cover a wide variety of applications, including autonomous cars maneuvering through complex road networks and industrial robots enhancing manufacturing procedures [19]. The ability of autonomous systems to operate in unstructured and uncertain environments while continuously learning and improving underscores their significance in modern technology landscapes [20]. Historically, the concept of autonomy in systems can be traced back to early developments in robotics and artificial intelligence. The roots of autonomous systems can be traced back to the early 20th century, with the advent of automated machinery and systems designed to perform specific tasks without direct human intervention. However, it was not until the latter half of the century, with advances in computing power and algorithms, that autonomous systems began to truly flourish [21]. One notable milestone in the development of autonomous systems was the introduction of the first industrial robots in the 1960s. These early robots were programmed to perform repetitive tasks on factory assembly lines, marking the beginning of automation in manufacturing processes. Over the decades, advancements in robotics led to the emergence of more sophisticated autonomous systems capable of operating in dynamic and unpredictable environments [22]. In the realm of transportation, the idea of autonomous vehicles dates back to the 1920s, with early experiments in automatic driving systems. However, it wasn't until the late 20th century that significant progress was made in this area, fueled by advancements in sensors, computing, and machine learning algorithms. The advancement of self-driving vehicles, especially, has attracted significant interest lately, with various companies and research organizations striving to make autonomous vehicle technology a reality [23]. Another critical application of autonomous technologies is in the development of Advanced Driver Assistance Systems (ADAS). ADAS encompasses a range of electronic systems that assist drivers in driving and parking functions, enhancing

TABLE I
EXISTING SURVEYS ON AUTONOMOUS CARS, ROBOTICS, DRONES, AND ADVANCED DRIVER ASSISTANCE SYSTEMS (ADAS) USING DRL

Ref.	Autonomous Cars	Autonomous Robotics	Autonomous Drones	ADAS	Applications	Main Achievement
[10]	X	✓	X	X	Indoor and crowd navigation	Demonstrates benefits and novel solutions of DRL in mobile robot navigation.
[11]	X	✓	X	X	Visual navigation for agents	Navigation for agents using deep reinforcement learning, classifying DRL visual navigation approaches and addressing challenges such as data inefficiency and limited generalization.
[12]	X	✓	X	X	Grasping, object handling	DRL in manipulation tasks for robotics, discussing principles, algorithms, obstacles, and solutions in deploying these algorithms.
[13]	✓	X	X	X	Various driving scenarios	Overview of RL in autonomous driving systems, addressing challenges of RL, validation, safety, open-source frameworks.
[14]	✓	X	X	X	Intersection management, urban navigation, highway motoring	Discusses DRL applications in autonomous driving scenarios and challenges in motion planning.
[15]	X	X	✓	X	Drone path planning, navigation through unknown environments	Provides insights into optimal policies and comparative performance of DQN and DDPG in drone navigation.
[16]	✓	X	X	X	Car-following, lane-keeping, merging, dense traffic navigation	DRL in the context of autonomous vehicle motion planning, discussing challenges, progress, and the importance of multi-agent reinforcement learning for interactions with other vehicles.
[17]	✓	X	X	X	Lateral (steering), longitudinal (acceleration and braking), simultaneous lateral and longitudinal control	Comprehensive survey of deep learning applications in autonomous vehicle control, emphasizing the need for extensive real-world testing and highlighting trends towards deeper models and temporal cues.
[18]	X	X	✓	X	Multi-drone systems coverage, adversarial search, computational offloading	Examines the application of DRL in multi-drone systems, focusing on scalability and cooperation without centralized control, highlighting its importance and providing a critical analysis of the current state of research.
[74]	X	X	X	✓	Vehicle detection in ADAS and autonomous vehicles	In-depth analysis of vehicle detection methods, evolution from traditional to deep learning approaches, including deep reinforcement learning (DRL), challenges in various environments, importance of efficient algorithms, superiority of OPMs, integration of DNNs, CNNs, and DRL techniques, experimental results with VEDAI database, insights into challenges of aerial vehicle detection.
Our Survey	✓	✓	✓	✓	Autonomous Systems: cars, robotics, drones, adas	A comprehensive overview of DRL applications in autonomous systems, identifies research gaps, and suggests future research directions.

vehicle safety and road safety in general. These systems use a combination of sensors, cameras, radar, and LIDAR to monitor the vehicle's surroundings and provide real-time data to assist in decision-making [24]. The history of ADAS can be traced back to the early implementations of anti-lock braking systems (ABS) in the 1980s, which prevented wheels from locking up during braking, maintaining traction and control. Following ABS, the introduction of features like electronic stability control (ESC) and adaptive cruise control (ACC) further advanced vehicle safety and driving comfort. Over time, ADAS has evolved to include sophisticated functionalities such as lane departure warning, collision avoidance, blind-spot detection, and automated parking systems [25]. These systems are designed to reduce human error, which is a significant factor in traffic accidents, by providing alerts or taking

control of the vehicle in critical situations. The integration of Deep Reinforcement Learning (DRL) into ADAS has the potential to enhance these systems further by enabling more adaptive, intelligent, and context-aware functionalities. DRL can improve the performance of ADAS by learning from vast amounts of driving data, allowing the systems to handle a wide variety of driving conditions and scenarios with greater precision and reliability [26]. Similarly, the use of autonomous drones has grown rapidly in various fields, including agriculture, surveillance, and logistics. Originally developed for military applications, drones have evolved into versatile tools capable of performing a wide range of tasks autonomously, thanks to advancements in sensor technology, flight control systems, and artificial intelligence [27]. Overall, the historical evolution of autonomous systems has been characterized by

TABLE II
LIST OF ACRONYMS

Acronym	Definition	Acronym	Definition
2LL-CACC	Two-Layer Linear Causal Adaptive Cruise Control	IRL	Inverse Reinforcement Learning
A2C	Advantage Actor-Critic	LDW	Lane Departure Warning
A3C	Asynchronous Advantage Actor-Critic	LLM	Large Language Model
ABS	Anti-lock Braking Systems	LSTM-RL	Long Short-Term Memory Reinforcement Learning
ACC	Adaptive Cruise Control	MARL	Multi-Agent Reinforcement Learning
ACKTR	Actor-Critic using Kronecker-Factored Trust Region	MEC	Multi-access Edge Computing
ADAS	Advanced Driver Assistance Systems	ML	Machine Learning
AI	Artificial Intelligence	MLP	Multi-Layer Perceptron
AV	Autonomous Vehicle	MPC	Model Predictive Control
AVS-RL	Adaptive Vector Scaling RL	NAF	Normalized Advantage Function
BVLOS	Beyond Visual Line of Sight	OPM	Object Proposal Method
CACC	Cooperative Adaptive Cruise Control	POMDP	Partially Observable Markov Decision Process
CADRL	Category-aware Dual-agent Reinforcement Learning	PP	Pure Pursuit
CAM-RL	Crowd Aware Memory-based Reinforcement Learning	PPO	Proximal Policy Optimization
CARLA	Car Learning to Act	RL	Reinforcement Learning
CAV	Connected Autonomous Vehicles	RPN	Region Proposal Network
CNN	Convolutional Neural Network	SAC	Soft Actor-Critic
COMAU SMART3-S2	A robotic manipulator COMAU's SMART series	SARL	Semi-Automatic Reinforcement Learning
D2QN	Double Deep Q Network	SDRL	Swarm Deep Reinforcement Learning
D3QN	Double Dueling Deep Q Network	SIFRCNN	Scale Invariant Faster Region-based Convolutional Neural Networks
D-A3C	Delayed Asynchronous Advantage Actor-Critic	SUMO	Simulation of Urban MObility
DDPG	Deep Deterministic Policy Gradient	TANet	Target Assignment Network
DNN	Deep Neural Networks	TD3	Twin-delayed Deep Deterministic policy gradient
DQN	Deep Q Networks	TD	Temporal Difference
DQN-C51	Categorical Deep Q Network with 51 distributions	TORC	The Open Racing Car Simulator
ESC	Electronic Stability Control	TPU	Tensor Processing Unit
GANs	Generative Adversarial Network	VEDAI	Vehicle Detection in Aerial Imagery
GPU	Graphics Processing Unit	VILDS	Vision-based Ingenious Lane Departure Warning System
GRU	Gated Recurrent Units	VRU	Vulnerable Road Users

a steady progression from rudimentary automated machinery to sophisticated (see Fig. 2), adaptive systems capable of operating in complex and dynamic environments. These advancements have paved the way for the integration of Deep Reinforcement Learning (DRL), further enhancing the autonomy, adaptability, and intelligence of modern autonomous systems.

B. Deep Reinforcement Learning

Understanding fundamental ideas and historical background is critical when researching Deep Reinforcement Learning (DRL) and its importance for autonomous systems. DRL combines deep learning with reinforcement learning approaches, allowing machines to acquire optimum decision-making strategies through trial and error in challenging environments. This method is critical for autonomous systems, which require agents to navigate dynamic and uncertain surroundings without human involvement [9]. The historical timeline of DRL highlights its progression from fundamental principles in reinforcement learning to ground-breaking advances in deep neural networks. Milestones such as DeepMind's launch of Deep Q-Networks (DQN) in 2013 transformed the field by proving the capability of learning control rules directly from raw sensory input. Along with DQN, later algorithms such as Deep Deterministic Policy Gradient (DDPG) and Proximal Policy Optimization (PPO) improved the stability and efficiency of DRL methods, allowing for their inclusion in a variety of autonomous systems [29]. Understanding terminology like policy, value function, and autonomous systems reveals the fundamental ideas that support DRL's applicability in

real-world scenarios. Understanding these important principles and historical developments allows academics to recognize DRL's transformational potential in determining the future of autonomous systems [30]. DRL has spawned a variety of algorithms, each designed for specific types of tasks and environments:

- *Deep Q-Network (DQN)* is a value-based reinforcement learning algorithm that integrates deep learning with Q-learning. The core idea is to use a deep neural network to approximate the Q-value function. The Q-value function quantifies the quality of different actions taken from a particular state. During training, experiences in the form of state, action, reward, and next state (s, a, r, s') are stored in a replay buffer [31]. The neural network is then trained on samples from this buffer to minimize the difference between the predicted Q-values and the target Q-values, which are computed using the Bellman equation. [32] This process helps in stabilizing the learning by breaking the correlation between consecutive observations making it effective in environments where agents learn from raw sensory data.
- *Deep Deterministic Policy Gradient (DDPG)* is an actor-critic algorithm that handles environments with continuous action spaces. The "actor" in DDPG proposes an action given a state, while the "critic" evaluates the action by computing the Q-value. The actor is updated using a policy gradient method aiming to maximize the Q-value provided by the critic [33]. Simultaneously, the critic is updated based on the Temporal Difference (TD) error, similar to Q-learning [34]. DDPG uses

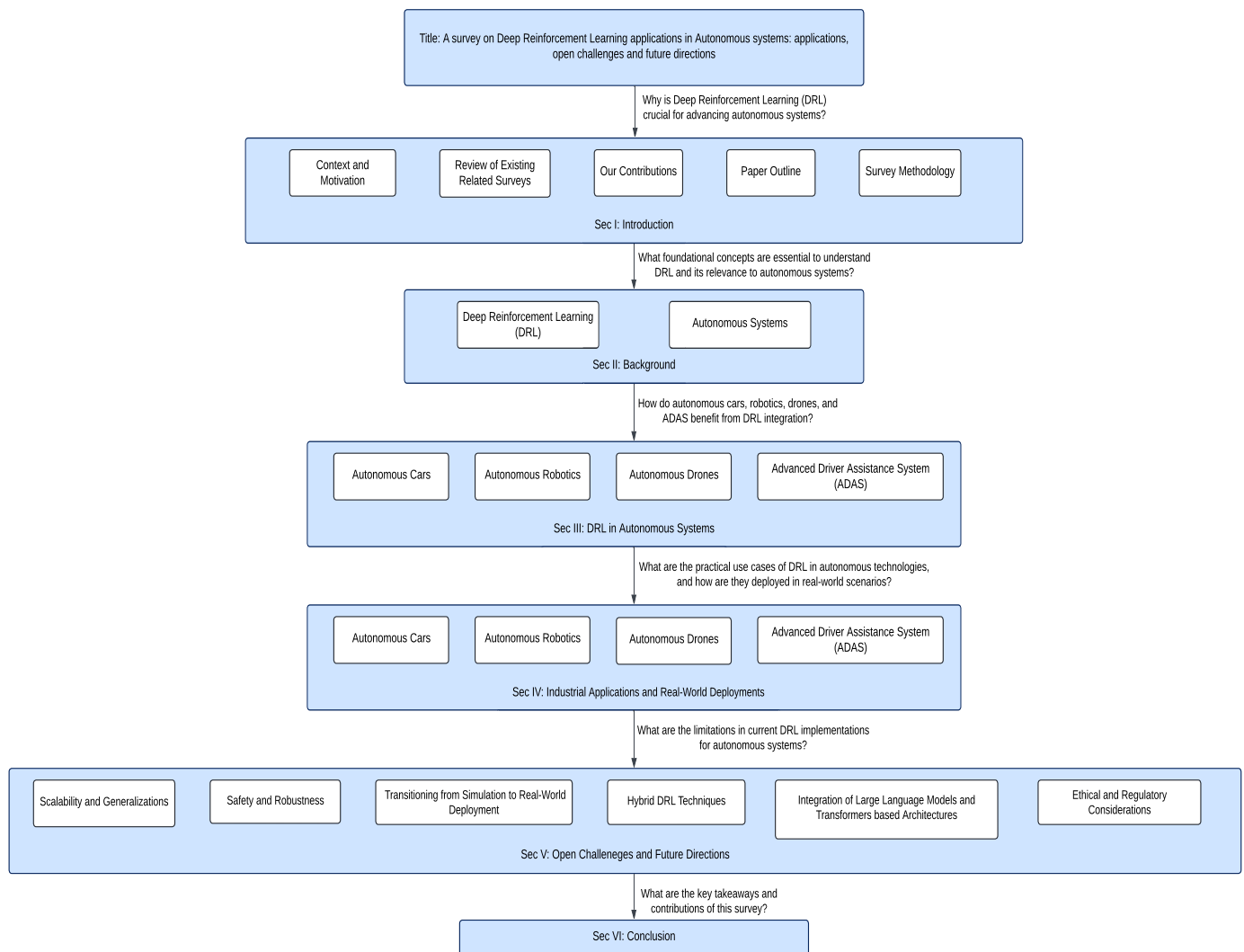


Fig. 1. The structure of the article.



Fig. 2. Evolution of autonomous systems [28].

two key techniques: experience replay for stabilizing training and target networks to compute stable target

values, reducing the risk of large updates destabilizing the learning process making it suitable for continuous action

spaces and is extensively used in applications requiring precise control, such as autonomous driving and robotic manipulation [33].

- *Proximal Policy Optimization (PPO)* is a policy gradient method that optimizes a “surrogate” objective function. This surrogate objective is designed to take into account the change in policy by incorporating a ratio of the new policy probability to the old policy probability, clipped to be within a specified range. This clipping mechanism effectively limits the size of the policy update, reducing the risk of harmful large updates that could destabilize the training process. PPO alternates between sampling data through interaction with the environment and optimizing the surrogate objective using stochastic gradient ascent making it ideal for a wide range of robotic applications [35].
- *Soft Actor-Critic (SAC)* is an off-policy actor-critic algorithm that introduces an entropy term to the reward function, promoting exploration by encouraging the actor to explore actions with uncertain outcomes. This entropy term is added to the traditional reward, thus not only maximizing the expected reward but also maximizing entropy. In practice, SAC learns both a policy (actor) that suggests actions and a Q-value function (critic) that evaluates actions, with the actor being trained to maximize a trade-off between expected return and entropy, fostering robust exploration and exploitation capabilities [36].
- *Delayed Asynchronous Advantage Actor Critic (D-A3C)* modifies the Asynchronous Advantage Actor Critic (A3C) by introducing a delay in the gradient updates to stabilize training. In standard A3C, multiple agents (or “workers”) independently interact with multiple instances of the environment and compute gradients. These gradients are then asynchronously applied to a central model [37]. In D-A3C, there is a deliberate delay introduced before applying these gradients to the central model. This delay helps in accumulating a more stable and reliable set of gradients, [38] reducing the variance and potential instabilities caused by very frequent and possibly conflicting updates from multiple workers making it beneficial in model-free reinforcement learning environments like autonomous driving.

III. DEEP REINFORCEMENT LEARNING IN AUTONOMOUS SYSTEMS

In this section, our focus is primarily center on the applications of DRL in autonomous cars, robotics, and drones. Additionally, we will present a comprehensive overview of recent research conducted in the past few years across these three autonomous systems.

A. Autonomous Cars

Several research papers delve into the application of Deep Reinforcement Learning (DRL) techniques across various aspects of autonomous driving, ranging from lane following and parking maneuvers to urban navigation and collision avoidance strategies.

1) *Lane Following and Parking Maneuvering*: This theme emphasizes allowing vehicles to adhere to lane discipline and execute accurate parking maneuvers. The challenges tackled include lateral control (remaining centered within the lane), longitudinal control (upholding suitable speed), and adapting to changing road conditions such as traffic, weather, and obstacles. These applications are essential for attaining smooth and safe autonomous driving in organized settings like highways and parking facilities. Both research papers [39] and [40] delve into the application of Deep Reinforcement Learning (DRL) for autonomous vehicle control in distinct scenarios. Paper [39] focuses on lane following, employing a Deep Deterministic Policy Gradient (DDPG) algorithm implemented in MATLAB and SIMULINK. It evaluates system performance using metrics like episode steps, rewards, and Q0 values across various driving scenarios, emphasizing lateral and longitudinal control in dynamic environments. The study underscores DRL’s role in enhancing autonomous driving safety and surpassing conventional controllers’ limitations, demonstrating effective navigation on curved roads. Meanwhile, paper [40] explores DRL’s application in autonomous parking maneuvers—parallel, angular, and perpendicular—using the same DDPG algorithm. It investigates adaptability in varied parking scenarios with obstacles, different initial conditions, and distances from parking spots. The research highlights the agent’s capability to generalize movements but identifies challenges such as simultaneous parking from both sides and specific obstacle configurations. Both studies contribute to advancing autonomous vehicle capabilities through DRL, addressing challenges in real-time navigation and precision parking tasks. (see Fig.3)

2) *Urban Navigation and Vehicle Routing*: The focus of DRL techniques in this area is to enhance navigation within complex urban settings. This includes efficiently guiding vehicles through busy regions, overseeing interactions with other cars, pedestrians, and traffic lights, and adjusting to unforeseen behaviors in diverse traffic situations. The objective is to minimize travel duration, boost fuel efficiency, and enhance overall traffic management. Recent advancements in Deep Reinforcement Learning (DRL) have significantly enhanced autonomous driving and vehicle routing strategies. In [42], a model-free Deep Reinforcement Learning Planner employs a delayed version of Asynchronous Advantage Actor Critic (D-A3C) to predict acceleration and steering angles, facilitating safe and comfortable urban driving. This approach emphasizes training in obstacle-free simulations and testing in real-world conditions, aided by a deep response model that minimizes the gap between simulation and reality. Additionally, pre-training with Imitation Learning improves path following and adherence to speed limits, showcasing robust generalization capabilities. Future directions include expanding to multi-agent systems and addressing challenges related to visual input limitations. On the other hand, [43] introduces a deep Q-network (DQN)-based multi-objective vehicle routing control for autonomous vehicles (AVs) in mixed traffic environments. This method optimizes traffic flow at intersections, demonstrating superior performance in driving distance, time efficiency, and AV waiting times compared to conventional methods. The study

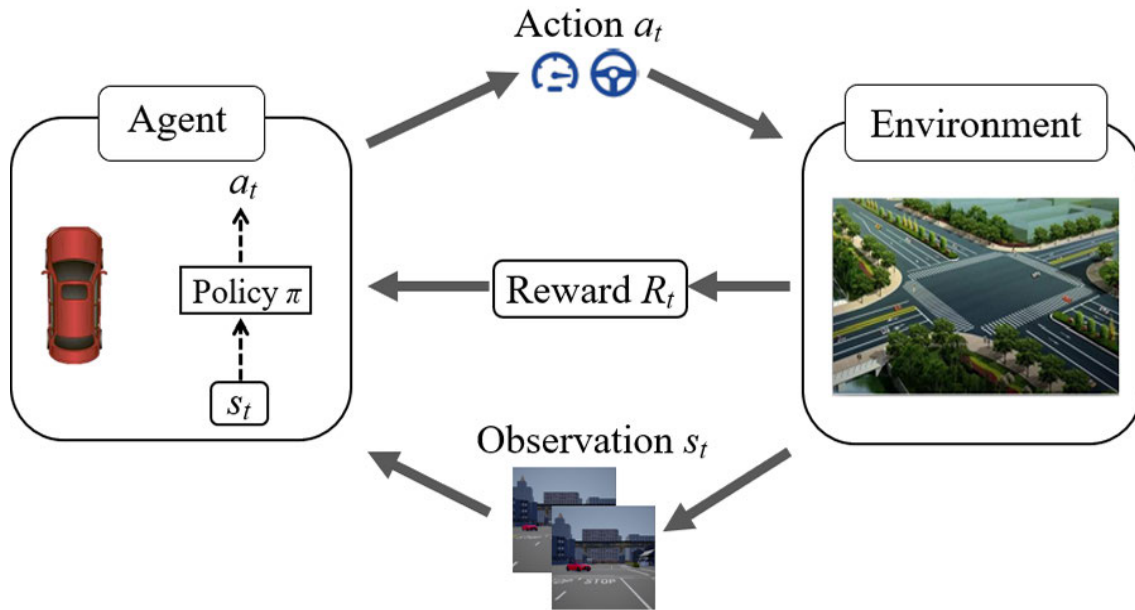


Fig. 3. DRL enabled decision-making for autonomous driving [41].

underscores the significance of efficient traffic distribution in dynamic scenarios and discusses future research avenues to enhance its applicability.

3) *Obstacle and Collision Avoidance*: Prioritizing safety is essential in autonomous driving, and this theme focuses on DRL applications to avoid collisions with both stationary and moving objects, such as other vehicles, pedestrians, or road barriers. Key challenges include real-time decision-making and planning in dynamic environments to prevent accidents while maintaining smooth navigation. Three distinct approaches to enhancing obstacle avoidance and collision mitigation in autonomous driving systems are explored in recent research. The study described in [44] investigates end-to-end self-driving models using reinforcement learning (RL) on the AWS DeepRacer platform. It evaluates various RL algorithms such as Soft Actor Critic (SAC) and Proximal Policy Optimization (PPO), focusing on sensor configurations, action spaces, and reward functions to optimize obstacle avoidance. Addressing challenges like Sim-to-Real transfer and real-world deployment, the research emphasizes the role of sensor selection and training duration in improving model performance. The findings underscore the importance of bridging the gap between simulated and actual environments for effective deployment of intelligent agents in real-world scenarios. In contrast, [26] proposes a hierarchical deep reinforcement learning (DRL) framework targeting collision avoidance, particularly emphasizing safety for vulnerable road users (VRUs). Utilizing Double Deep Q-Network (D2QN), the framework integrates a PID-based pure pursuit controller for routine navigation and activates DRL for dynamic obstacles like pedestrians in complex traffic scenarios simulated in CARLA. This approach demonstrates D2QN's effectiveness in reducing action value overestimation inherent in traditional DQN models, thereby advancing the safety and reliability of autonomous driving systems in interacting with VRUs. Additionally, paper [45] presents a DRL approach for autonomous vehicle navigation,

focusing on lane keeping, overtaking, and collision avoidance. Using a deep deterministic policy gradient (DDPG) algorithm, the study trains an autonomous vehicle agent within the TORCS simulator. The agent utilizes multi-modal sensor inputs to perceive its environment and make sequential decisions for safe navigation. The study demonstrates the effectiveness of the DDPG technique in various simulated driving scenarios, addressing complex driving maneuvers and safety challenges, and contributing to the advancement of autonomous driving technology.

4) *Braking System*: Autonomous braking systems powered by DRL are critical for ensuring safety during emergencies. This theme explores the use of DRL to model and execute timely braking actions, minimizing the risk of accidents in scenarios like sudden pedestrian crossings, abrupt vehicle stops, or adverse weather conditions. Two recent papers explore the application of deep reinforcement learning (DRL) techniques in autonomous braking systems. Paper [46] focuses on a deep Q-network (DQN) approach, aiming to autonomously control vehicle braking based on real-time sensor data detecting potential collisions with obstacles such as pedestrians. It introduces a reward function that optimizes between minimizing collision damage and ensuring efficient braking, supplemented by a "trauma memory" mechanism to enhance learning from rare events like collisions. The effectiveness of this approach is demonstrated through simulations, including successful performance in Euro NCAP autonomous emergency braking tests, highlighting DRL's potential in enhancing autonomous vehicle safety. On the other hand, paper [47] proposes a deep deterministic policy gradient (DDPG) algorithm for autonomous braking decisions in emergencies. This method employs an actor-critic framework to balance safety, efficiency, and passenger comfort, showcasing superior performance over traditional methods in extensive simulations. By introducing a multi-objective reward function, it enhances learning effectiveness and decision-making accuracy, thereby contributing

significantly to the safety of connected and autonomous vehicles (CAVs) during critical scenarios.

5) *Comparison and Analysis*: The advancements in autonomous driving systems can be attributed to the continuous evolution of DRL techniques. The studies reviewed demonstrate significant progress across various facets of autonomous vehicle (AV) capabilities, each employing distinct DRL approaches tailored to specific tasks such as lane following, parking maneuvers, urban driving, obstacle avoidance, and collision avoidance. As given in the Table III. Below, is the comparative analysis based on the key findings from each study, highlighting their contributions, experimental setups, and outcomes:

- *DRL Techniques and Objectives*: The surveyed papers predominantly utilize Deep Deterministic Policy Gradients (DDPG) and Deep Q-Network (DQN) variants, with notable applications of Double Deep Q-Network (D2QN) and Soft Actor-Critic (SAC). These techniques have been deployed to address diverse objectives. For instance, lane following and lateral/longitudinal control [39] employed DDPG, enabling systems to achieve autonomous driving in dynamic environments while effectively maneuvering along curved roads. Similarly, parking maneuvers [40] demonstrated the adaptability of DDPG in executing parallel, angular, and perpendicular parking across various scenarios. Urban driving and obstacle avoidance [42] utilized D-A3C to develop a planner ensuring safe urban navigation, supported by both real-world and simulation experiments. In the context of vehicle routing and mixed traffic control [43], DQN optimized traffic flow and reduced travel time in simulated urban environments. For collision avoidance for vulnerable road users (VRUs) [26], D2QN integrated with a PID-based pure pursuit controller handled complex traffic scenarios effectively.
- *Experiment Setup and Testing Methods*: The experimental setups across these studies vary widely, encompassing simulators like MATLAB, SIMULINK, and CoppeliaSim, as well as real-world testing environments. Each study's experimental design was crucial in validating the performance of the proposed DRL techniques. Simulators, such as the CARLA simulator [26], TORCS [45], and AWS DeepRacer platform [44], created controlled yet challenging scenarios for testing DRL algorithms. Testing methods included metrics like episode steps, rewards, and Q-values, reflecting the learning efficiency and decision-making capabilities of the agents. Comprehensive evaluations were achieved through both real-world and synthetic experiments [42], [44].
- *Results and Effectiveness*: The effectiveness of DRL-based autonomous driving systems was evident across various applications. Systems showcased the ability to navigate complex road structures and adapt to dynamic environments, enhancing safety and performance [39], [45]. High adaptability and successful execution of parking maneuvers were notable outcomes, reflecting the robustness of DRL in handling diverse driving tasks [40]. Optimized traffic flow and reduced travel time in urban and mixed traffic scenarios underscored the potential of

DRL in managing real-world urban traffic conditions [43]. Moreover, DRL techniques significantly improved safety, especially in collision avoidance scenarios for VRUs and autonomous braking systems [26], [46], [47].

B. Autonomous Robotics

Several papers delve into the application of Deep Reinforcement Learning (DRL) across various domains of robotics, showcasing its versatility and efficacy.

1) *Swarm Behaviour*: Swarm robotics emphasizes collective behavior, where multiple robots work collaboratively to complete tasks. This theme addresses coordination, adaptability, and resilience in dynamic and uncertain environments, such as search-and-rescue missions or industrial automation. DRL methods optimize decision-making across the swarm to ensure efficiency and robustness. Two distinct methodologies harnessing deep reinforcement learning for swarm robotics are examined in recent research. The first study [48] explores the application of proximal policy optimization (PPO) to train controllers for a multi-legged robotic swarm operating on flat and rough terrains. Through simulations, the PPO algorithm successfully generates collective behaviors such as walking and formation into lines, showcasing robust performance across varying terrain conditions. The research emphasizes the effectiveness of deep reinforcement learning in optimizing swarm behavior through specific task-oriented training. In contrast, the second paper [50] introduces Swarm Deep Reinforcement Learning (SDRL), a decentralized framework integrating blockchain technology with deep reinforcement learning for robotic manipulation tasks. SDRL enables multiple agents to autonomously control local robots while maintaining privacy and leveraging a blockchain network for parameter commitment. The study demonstrates that SDRL enhances model speed and accuracy, particularly benefiting from increased learning efficiency with additional computational resources. By combining deep learning techniques with decentralized control mechanisms, SDRL represents a significant advancement in enabling adaptive and collaborative behaviors among robotic swarms.

2) *Collision Avoidance*: DRL techniques in this subtopic are designed to prevent collisions during robotic navigation, especially in environments with static or moving obstacles. Applications include industrial robots operating in confined spaces or service robots interacting with humans, focusing on safety and operational continuity. Two recent studies propose innovative approaches to collision avoidance in robotics, each focusing on distinct applications and methodologies. Paper [53] introduces the Crowd Aware Memory-based Reinforcement Learning (CAM-RL) method, emphasizing collision-free navigation in crowded environments by leveraging gated recurrent units (GRU) to model human dependencies and behaviors. This method integrates multilayer perceptrons and attention mechanisms to enhance human-robot interactions, demonstrating superior performance compared to existing techniques like CADRL, LSTM-RL, and SARL in scenarios with varying numbers of humans. The study underscores the complexity of navigating in crowded spaces with

TABLE III
COMPARISON OF DRL TECHNIQUES IN AUTONOMOUS DRIVING

Ref, Year	Objective	DRL Technique	Experiment Setup	Performance Metrics	Results
[26], 2024	Collision avoidance for VRUs	Double Deep Q-Network (D2QN)	CARLA simulator	- Collision Avoidance Success Rate: 95% - Avg. Response Time: 0.5 seconds - Computational Efficiency: 30ms per decision cycle - Training Time: 10 hours - Accuracy Gain: 15% (VRU detection) - Resource Efficiency: 20% reduction in computational usage	- Effective handling of complex traffic scenarios. - Improved safety for vulnerable road users.
[39], 2023	Lane following, lateral and longitudinal control (Straight & Curved Roads)	Deep Deterministic Policy Gradients (DDPG)	MATLAB, SIMULINK	- Max Reward: 5000 (Straight), 4800 (Curved) - Avg. Reward: 4600 (Straight), 4700 (Curved) - Training Efficiency: Transition efficiency between scenarios achieved in 100 episodes. - Lateral Deviation: Maintained within ± 0.5 m	Effective lane-keeping and training efficiency improvement by transferring knowledge between scenarios.
[40], 2022	Parking maneuvers (parallel, angular, perpendicular)	Deep Deterministic Policy Gradients (DDPG)	CoppeliaSim	- Avg. Reward: 194.1 (Obstacle-free), 193.9 (With obstacles) - Success Rate: High across all parking types(not specified). - Training Episodes: 3000 - Testing Episodes: 50 for each scenario - Adaptability: Effective generalization to various environments	Minimal performance impact in obstacle environments; challenges in parking from both sides of parking spots.
[42], 2022	Urban driving, obstacle avoidance	D-A3C	Synthetic simulator with HD Maps	- Training time reduced via pre-training with Imitation Learning - Success rate: 100% in simulation - Real-world success with Policy 2 (handling dynamics and following rules)	Demonstrated efficiency improvements for traffic flow and route optimization. Generalization capability observed; effective for low-traffic environments with scalability challenges noted for future work.
[43], 2024	Vehicle routing control in mixed traffic	Deep Q-Network (DQN)	SUMO, City1 and City2 maps	- Avg. Driving Speed: 7.55 m/s (City1), 8.89 m/s (City2) - Driving Distance: 12–25% improvement (City1), 7–15% (City2) - Waiting Time Reduction: 18–28% (City1), 7–17% (City2) - Neural Network: Four layers (150 and 100 neurons in hidden layers)	Improved traffic flow, higher average speeds, and effective traffic distribution.
[44], 2024	Obstacle avoidance	Various (SAC, PPO)	AWS DeepRacer platform	- Collision Rate Improvement: SAC (95.8%), -Completion Time Reduction: SAC (79%).	SAC with LiDAR performed best across metrics. Continuous reward functions showed potential for improved learning outcomes. - Domain adaptation identified for Sim-to-Real transfer.
[45], 2023	Lane keeping, overtaking, collision avoidance	Deep Deterministic Policy Gradient (DDPG)	TORCS simulator	- Cumulative Reward: 1500 - Lane-Keeping Accuracy: 95% - Overtaking Success: 90% - Collision Avoidance Rate: 85% - Training Episodes: 2000 - Avg. Time/Episode: 0.5 seconds	- Effective handling of complex driving maneuvers. - Contributions to safety and autonomous driving technology.
[46], 2017	Autonomous braking system	Deep Q-Network (DQN)	Computer simulations (Euro NCAP autonomous emergency braking tests)	- Collision avoidance rate: Over 90% - Response time: 0.5 seconds	- Real-time decision-making for braking. - Improved safety with trauma memory mechanism. - Successful performance in safety tests.
[47], 2020	Autonomous braking in emergencies	Deep Deterministic Policy Gradient (DDPG)	Extensive simulations	- Collision Rate: 1.43% (Proposed Method), 5.86% (DDPG w/o Steering), 13.44% (DQN w/ Discrete Deceleration) - Comfort Level: Exceeding comfort limits at 9.21% (Proposed), 25.48% (DDPG w/o Steering), 42.64% (DQN w/ Discrete Deceleration) - Reward Stability: Achieved within 400 episodes	- Superior performance over traditional methods. - Enhanced safety, efficiency, and passenger comfort, highlighting efficiency in handling emergency scenarios.

diverse human behaviors and velocities, aiming to improve overall robot performance through comprehensive understanding and efficient adaptation. In contrast, paper [51] focuses on collision avoidance for robotic manipulators using a deep reinforcement learning (DRL) approach based on the Normalized Advantage Function (NAF) algorithm. This model-free method addresses continuous space challenges inherent in robotic systems, particularly in ensuring safe human-robot interactions with unpredictable obstacles. Demonstrated with a COMAU SMART3-S2 manipulator, the study illustrates the algorithm's effectiveness in real-time simulations, highlighting its adaptability and suitability for complex environments requiring agile and safe robotic navigation strategies.

3) *Trajectory Planning*: Trajectory planning involves determining optimal paths for robotic movement while avoiding obstacles and minimizing energy or time consumption. DRL methods optimize navigation under uncertain or constrained conditions, such as crowded spaces or environments with limited sensory data. Several recent studies explore the integration of Deep Reinforcement Learning (DRL) techniques with traditional control methods for trajectory planning in robotics. Paper [49] introduces a hybrid approach for path-following control in non-holonomic mobile robots, combining steering control with the Soft Actor-Critic (SAC) algorithm. The method adapts robot velocities dynamically, demonstrating superior performance in reducing cross-track error and enhancing path-following success rates compared to traditional Pure Pursuit (PP) control with constant speed. However, discrepancies between simulation and real-world experiments highlight challenges related to control dynamics and input-output delays, suggesting areas for future improvement. In contrast, paper [52] focuses on autonomous robot navigation using DRL, emphasizing safety and compliance in dynamic environments. Employing the deep deterministic policy gradient (DDPG) algorithm, the study showcases successful maneuvers like obstacle avoidance and lane keeping, indicating DRL's effectiveness in handling non-holonomic constraints and complex driving scenarios. Both papers underscore the potential of DRL in enhancing trajectory planning for robots operating under diverse environmental and dynamic constraints.

Additionally, paper [54] investigates DRL's application specifically in trajectory planning for robot manipulators in uncertain environments. The study compares Soft Actor-Critic (SAC) and Deep Deterministic Policy Gradient (DDPG) algorithms for motion planning of a Panda robot manipulator in dynamic scenarios. SAC emerges as more stable and consistent, achieving a 100 safety rate through effective collision avoidance within 6,000 simulation episodes. This research underscores SAC's capability in optimizing trajectory planning by ensuring both safety and accuracy, thereby suggesting promising avenues for integrating DRL into real-world robotic applications.

Collectively, these papers showcase the breadth of DRL applications in robotics, spanning diverse domains such as swarm behavior, collision avoidance, and trajectory planning. Research highlights include the use of PPO and SDRL for optimizing collective behaviors in multi-legged robotic swarms

across different terrains, demonstrating robust performance and decentralized control capabilities. Furthermore, advancements in collision avoidance techniques leverage models like CAM-RL and NAF to enhance robot navigation in crowded and dynamic environments, addressing complexities posed by human interactions and unpredictable obstacles. Additionally, studies employing algorithms such as SAC and DDPG underscore DRL's effectiveness in trajectory planning for both mobile robots and manipulators, emphasizing safety, adaptability, and improved performance in real-world scenarios. (see Fig.4)

4) *Comparison and Analysis*: The integration of Deep Reinforcement Learning (DRL) techniques in autonomous robotics has significantly enhanced the capabilities of robots to navigate, manipulate, and perform complex tasks in dynamic and uncertain environments. This section provides a comparative analysis of various DRL techniques applied to autonomous robotics, focusing on their objectives, experimental setups, testing methods, and outcomes: (as given in Table IV)

- *DRL Techniques and Objectives*: The objectives of the studies vary, reflecting the diverse applications of DRL in autonomous robotics. For instance, [48] focuses on collective behavior in multi-legged robotic swarms using Proximal Policy Optimization (PPO), demonstrating its adaptability in rough terrains. On the other hand, [49] addresses path following for nonholonomic mobile robots by combining Soft Actor-Critic (SAC) with traditional steering control, achieving significant improvements in failure and completion rates. Robotic manipulation in dynamic environments is explored in [50] using Swarm Deep Reinforcement Learning (SDRL), which enhances manipulation quality. Collision avoidance, a critical aspect of autonomous robotics, is studied in [51] and [53]. The former uses Normalized Advantage Function (NAF) to navigate robotic manipulators around unpredictable obstacles, while the latter employs a combination of GRU, MLP, Attention, and CAM-RL to outperform existing methods in crowded environments. Autonomous navigation in dynamic environments is tackled by [52] using Deep Deterministic Policy Gradient (DDPG), achieving superior performance in obstacle avoidance and reducing travel time. Finally, trajectory planning in uncertain environments is addressed by [54], comparing SAC and DDPG, with SAC demonstrating superior stability, consistency, and collision avoidance.
- *Experiment Setup and Testing Methods*: The experimental setups and testing methods vary significantly across the studies. [48] uses the PyBullet physics engine for simulations, demonstrating PPO's effectiveness in designing controllers for multi-legged robotic swarms. In contrast, [49] employs the Gmapping algorithm to test path-following capabilities through metrics such as completion rate, crashes, and off-track penalties. [50] uses Ethereum Smart Contracts for robotic manipulation, leveraging observations from a camera and joint control commands. [51] simulates collision avoidance in the COMAU SMART3-S2 robot manipulator, while [52] and [54] utilize PyBullet and CoppeliaSim for their

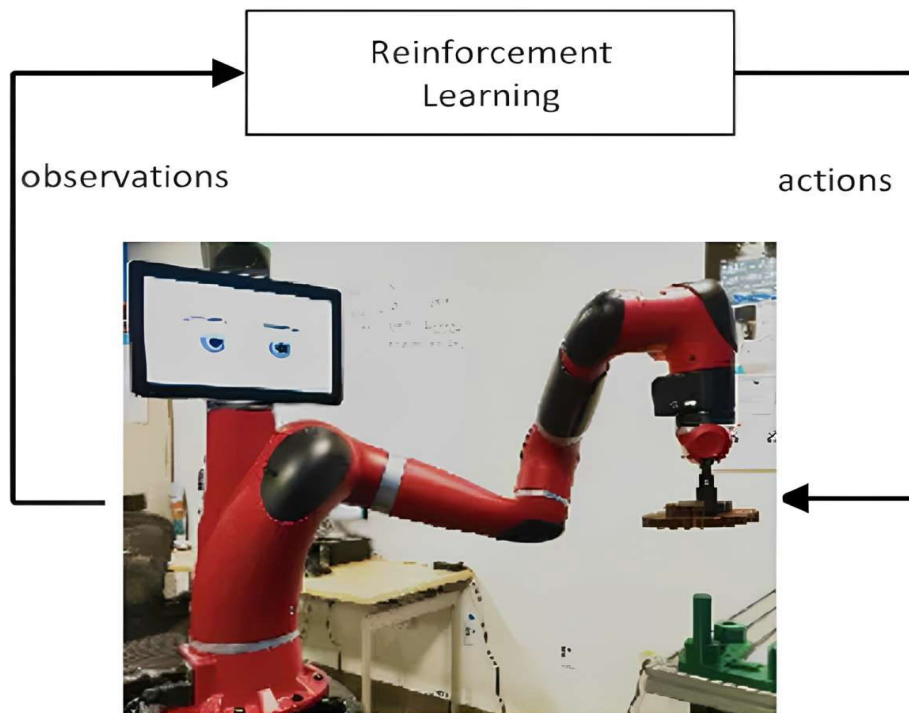


Fig. 4. DRL loop for industrial robot control [55].

respective experiments, focusing on obstacle avoidance and trajectory planning. [53] tests collision-free navigation in simulated crowded environments.

- *Results and Effectiveness:* The results indicate significant advancements in the application of DRL to autonomous robotics. PPO in [48] successfully designs controllers for multi-legged robotic swarms, showing higher adaptability in rough terrains. SAC combined with traditional steering control in [49] significantly improves failure and completion rates compared to constant speed control. In dynamic and complex environments, SDRL in [50] enhances robotic manipulation quality, improving speed and accuracy. NAF in [51] demonstrates real-time adaptability for collision avoidance in robotic manipulators. DDPG in [52] outperforms the dynamic window approach by adapting robot velocity to dynamic obstacles and reducing travel time. CAM-RL in [53] achieves higher success rates in crowded environments, outperforming existing methods. Lastly, SAC outperforms DDPG in [54] by achieving a 100% safety rate within 6,000 episodes, highlighting its stability and consistency in trajectory planning.

C. Autonomous Drones

The surge in autonomous drones, aka drone research, is now heavily influenced by the integration of Deep Reinforcement Learning (DRL) techniques, propelling advancements in autonomous capabilities across diverse tasks (see Fig.5).

1) *Security and Pollution Control:* The integration of advanced technologies in drones is pivotal for both security and pollution control. DRL applications in drones for security include tasks such as surveillance, intrusion detection,

and patrolling sensitive areas. For pollution control, drones equipped with sensors monitor and analyze air quality or locate and track pollution plumes in real-time, leveraging their mobility and coverage capabilities. The first paper [57] proposes a novel architecture for drone interception that leverages a self-navigation and supervised, model-free deep learning reinforcement approach. Utilizing deep reinforcement learning with cascade rewards, this method enables a single drone to navigate obstacles and intercept targets using only a forward-facing depth-RGB camera. The paper's key contributions include addressing the challenge of partial observability, optimizing the task of maximizing the overall expected reward, and developing a software library for training drones in target tracking and interception. Experimental results show that this approach surpasses state-of-the-art deep Q-network algorithms in learning policies, highlighting its potential for enhancing airport security, effective airspace patrolling, and safeguarding public and personal privacy. Complementing the security applications, the paper [58] presents a Multi-Agent Reinforcement Learning (MARL) scheme aimed at optimizing the mobility of a fleet of drones for characterizing air pollution plumes. This framework integrates pollutant concentration mapping using Gaussian Processes (GP) with trajectory planning via Deep Reinforcement Learning (DRL). Drones collect measurements to update the GP model and plan optimal trajectories, incorporating a weighted forgetting mechanism to adapt to temporal variations in pollution. Validated through extensive simulations, the proposed solution demonstrates rapid convergence, scalability, and adaptability to parameter changes. The findings contribute to timely pollution plume characterization, adaptive estimation of GP parameters, and the design of an IQL-based MARL model for drone trajectory planning.

TABLE IV
COMPARISON OF DRL TECHNIQUES IN AUTONOMOUS ROBOTICS

Ref, Year	Objective	DRL Technique	Experiment Setup	Performance Metrics	Results
[48], 2023	Collective behavior in multi-legged robotic swarm	Proximal Policy Optimization (PPO)	PyBullet physics engine	- Cumulative rewards (r1) - NLargest Cluster - Mean x-coordinate ($\bar{x}$) - dMajor and dMinor	- PPO successfully designed controllers for multi-legged robotic swarm. - Setting D (reward configuration) showed best performance in maintaining line formations - Higher adaptability in rough terrains. - 10 learning trials per reward setting, 50 re-evaluations
[49], 2024	Path following for nonholonomic mobile robots	SAC combined with traditional steering control	Gmapping algorithm	Failure rate: 0% - Completion rate: 100% - Average cross-track error: 0.05 m - RMSE: 0.07 m - Maximum error: 0.1 m - Average velocity: 0.5 m/s	- Significant improvement compared to Pure Pursuit (PP) control. - PP had higher failure rate, lower completion rate. - PP average cross-track error: 0.15 m, RMSE: 0.12 m. - 2000 training episodes
[50], 2022	Robotic manipulation in dynamic and complex environments	Swarm Deep Reinforcement Learning (SDRL)	Ethereum Smart Contracts	- Learning speed improvement: 32% increase with 3 workers vs 2 - Number of episodes to convergence: 300 for SDRL	- Improved speed and accuracy - Decentralized approach eliminates centralized parameter server - Tested with 1 to 8 workers - Used MoJoCo robot simulators with three-jointed robotic arms
[51], 2018	Collision avoidance in robotic manipulators	Normalized Advantage Function (NAF)	SCOMAU SMART3-S2 robot manipulator	- Obstacle distance - Total reward - Average loss	- Successfully navigated around unpredictable obstacles - Provided real-time, adaptable solution for collision avoidance - Time step: 50 ms per decision - 360 time steps in total
[52], 2023	Autonomous robot navigation in dynamic environments	Deep Deterministic Policy Gradient (DDPG)	CoppeliaSim	- Learning time: 6.57 hours - Maximum legal linear speed: 0.2 m/s - Comfort levels: 0.5 m/s ² (longitudinal), 1 rad/s ² (angular)	- Outperformed dynamic window approach (DWA) - Successfully completed maneuvers: obstacle avoidance, overtaking, follow-up, lane keeping, and lane change - Generalized to new and unseen scenarios
[53], 2022	Collision-free robot navigation in crowded environments	GRU, MLP, Attention, CAM-RL	Simulated crowded environment	- Success Rate (SR): 0.99 (circular), 1.00 (square) - Collision Rate (CR): 0.01 - Average Time (μ T): 19.31s (circular), 16.78s (square) - Average Distance (μ D): 18.54m (circular), 16.18m (square)	- Outperformed CADRL, LSTM-RL, and SARL - Statistically significant improvement (p-value < 0.001) - Tested in circular and square crossing scenarios with ten humans
[54], 2022	Trajectory planning of robot manipulators in uncertain environments	SAC, Deep Deterministic Policy Gradient (DDPG)	PyBullet physics engine	- Safety Rate: 100% for SAC within 6,000 episodes - Accuracy: Higher for SAC (specific values not stated) - Convergence speed: Faster for SAC	- SAC outperformed DDPG in stability, consistency, and collision avoidance - DDPG showed variable accuracy and did not guarantee collision avoidance even after 10,000 episodes - SAC demonstrated better computational efficiency and learning dynamics

2) *Stability and Trajectory Planning*: Ensuring stability during operations is a critical challenge for drones, especially in environments with high wind or other external disturbances. This theme focuses on DRL-based techniques for maintaining stability and planning efficient and safe trajectories, enabling drones to perform tasks like package delivery or mapping

effectively. Several recent studies explore the application of deep reinforcement learning (DRL) in enhancing stability and trajectory and path planning for drones across various scenarios. Paper [59] introduces a hybrid architecture integrating a nominal controller with a robust policy learned through DRL, aiming to ensure stable control for tasks like

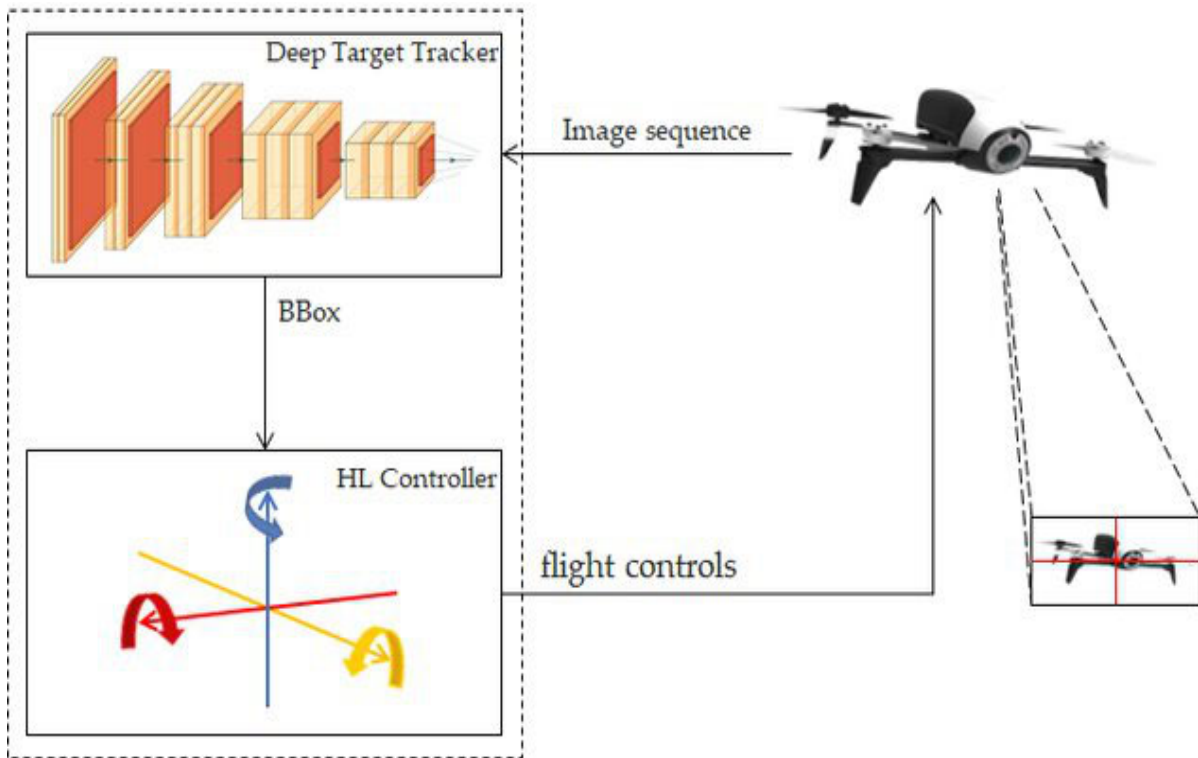


Fig. 5. Drone tracking framework using DRL [56].

wall-climbing drones. By employing an uncertainty-aware control mixer, the framework combines the stability of the nominal controller with the adaptive capabilities of the learned policy, validated through experiments with a tilting-hexarotor drone. Similarly, paper [60] proposes a DRL-based trajectory planning algorithm within a Partially Observable Markov Decision Process (POMDP) framework to optimize data collection and wireless power transfer in drone networks. This approach addresses challenges such as incomplete network state information due to environmental factors, demonstrating significant reductions in packet loss through simulations and indicating potential applications in dynamic network conditions.

Furthermore, paper [61] introduces an improved D3QN algorithm for autonomous path planning, highlighting its effectiveness in generating safe and feasible paths while avoiding obstacles in static and dynamic environments. This study contributes to optimizing path length and planning time compared to traditional algorithms, emphasizing the potential of reinforcement learning in overcoming complex path planning challenges. Similarly, paper [62] presents a DRL approach using Dueling Double Deep Q-networks (D3QN) for real-time path planning in dynamic and threat-laden environments. By integrating heuristic search rules with ϵ -greedy strategies, the framework assesses global situations to maximize drone survival probabilities, showcasing superior learning performance and stability over conventional DQN and D2QN algorithms. In addition, paper [63] investigates target assignment and path planning across multiple drones, offering a comprehensive formulation and simulation-based

validation of their proposed method. This study contributes to understanding the complexities of coordinating drones in various applications, laying groundwork for future advancements in multi-agent systems. These papers collectively advance the field by integrating advanced algorithms with practical applications, fostering robust solutions for stability, efficient trajectory planning, and coordinated multi-drone operations in diverse operational environments.

3) *Collision Avoidance*: Collision avoidance is essential for safe drone operation, particularly in busy or dynamic environments like urban areas or forests. DRL methods are used to navigate drones around obstacles in real-time while maintaining mission objectives. Three recent studies have explored Deep Reinforcement Learning (DRL) approaches for enhancing drone trajectory planning in complex environments. Three recent studies have explored Deep Reinforcement Learning (DRL) approaches for enhancing drone trajectory planning in complex environments. The papers [64] and [65] focus on deep reinforcement learning (DRL) approaches for collision avoidance in drones. In [64], the authors propose PICA, a probabilistic model, and RELIANCE, leveraging Deep Q-Networks (DQNs) to enable real-time decision-making for collision avoidance, adaptable to both drones and multi-access edge computing (MEC) facilities. RELIANCE demonstrates superior performance in simulations, showcasing the efficacy of DRL in enabling drones to learn from their environments. Similarly, [65] introduces a DRL framework using the Deep Deterministic Policy Gradient (DDPG) algorithm, incorporating soft and hard margin reward functions to facilitate smooth flight and reduce collision risks. This approach is validated

through experiments in simulated indoor environments, highlighting its potential for real-world applications.

Additionally, [66] explores obstacle avoidance using DRL in both discrete and continuous action spaces, integrating a U-net based segmentation model with an actor-critic network. Their findings indicate that continuous action space algorithms, especially ACKTR, outperform discrete ones and achieve performance on par with expert human pilots in drone racing scenarios. This study underscores DRL's capability to train drones for autonomous navigation in complex environments, minimizing the need for manual data labeling and enhancing the drones' navigation skills. Together, these studies illustrate the significant advancements in DRL techniques, demonstrating their potential to enhance the safety and efficiency of drone operations in diverse environments.

4) *Comparison and Analysis*: This section provides a comparative analysis of the recent advancements in the field of autonomous drones, specifically focusing on the application of Deep Reinforcement Learning (DRL) techniques. The provided Table V includes various studies that demonstrate the efficacy of different DRL methods in enhancing the performance of autonomous drones in diverse scenarios.

- *DRL Techniques and Objectives*: The primary objectives of these studies range from drone interception for airport security [57], air pollution plume characterization [58], stable control for wall-climbing drones [59], real-time trajectory planning [60], to collision avoidance [65] and obstacle avoidance [66]. The DRL techniques employed vary widely, showcasing the versatility and adaptability of DRL in addressing different challenges: Dueling Double DQN is used in both drone interception [57] and obstacle avoidance [66], highlighting its robustness in dealing with complex, dynamic environments. Multi-Agent Reinforcement Learning (MARL) combined with Gaussian Processes (GP) and Categorical Deep Q-Network (DQN-C51) is applied for air pollution plume characterization [58], illustrating the effectiveness of combining DRL with other machine learning techniques for environmental monitoring. Hybrid architectures with model-free DRL and Deep Q-networks (DQN) are utilized for stable control in wall-climbing drones [59] and real-time trajectory planning [60], respectively, demonstrating the ability of DRL to achieve stable and efficient control in both simulated and real-world scenarios. Twin-delayed deep deterministic policy gradient (TD3) with target assignment network (TANet) is implemented for multi-drone target assignment and path planning [63], showing advanced methods for optimizing resource allocation and navigation in multi-drone systems.
- *Experiment Setup and Testing Methods*: The studies utilize a variety of simulation platforms and experimental setups to validate their approaches. AirSim and Unreal Engine are employed for testing drone interception techniques [57], while NVIDIA Isaac Gym is used for real-world and simulation experiments for wall-climbing drones [59]. Python simulations using Google TensorFlow are applied for real-time trajectory planning [60], indicating the flexibility of DRL frameworks in adapting

to different simulation environments. Rectangular mission areas and adversarial obstacles, in combination with TensorFlow and Adam optimization, are used for path planning experiments [61], emphasizing the importance of creating realistic and challenging test scenarios. Simulated environments with varying complexity are employed for vision-guided collision avoidance [65], highlighting the need for scalable and generalizable solutions.

- *Results and Effectiveness*: The results from these studies underscore the significant improvements achieved through DRL techniques. The D3QN method demonstrates superior performance in both drone interception [57] and obstacle avoidance [66], outperforming state-of-the-art algorithms and achieving successful navigation in complex environments. The MARL scheme for air pollution plume characterization [58] optimizes drone fleet operations, achieving accurate estimations and rapid convergence, showcasing the potential of multi-agent systems in environmental monitoring. The Hybrid DRL framework for wall-climbing drones [59] ensures stability and improved performance in pose tracking and trajectory tracking, validating the effectiveness of integrating model-free DRL with traditional control methods. The Deep Q-network in a POMDP framework for real-time trajectory planning [60] significantly reduces packet loss and improves energy transfer efficiency, demonstrating the adaptability of DRL to dynamic network conditions. Additionally, the TD3 with TANet for multi-drone systems [63] achieves optimal target allocation and collision-free paths, outperforming existing methods and showcasing superior performance in dynamic and mixed settings.

D. Advanced Driver Assistance Systems (ADAS)

There are several research papers delve into the application of Deep Reinforcement Learning (DRL) techniques across various aspects and highlights significant advancements in Advanced Driver Assistance Systems (ADAS), ranging from lane departure warning and pedestrian detection to collision avoidance and adaptive cruise control strategies. These studies showcase how DRL can enhance safety, efficiency, and adaptability in ADAS.

1) *Lane Departure Warning/Correction*: This theme focuses on enhancing driver safety by detecting potential lane departures and providing timely warnings or corrective actions. DRL techniques are employed to predict lane deviations and control steering to maintain the vehicle within its designated lane. The Vision-based Ingenious Lane Departure Warning System (VILDS) and ADAS-RL represent significant advancements in lane departure warning systems for autonomous vehicles. The VILDS, as presented in [67], integrates Generative Adversarial Networks (GANs) and Long Short-Term Memory (LSTM) networks to enhance lane detection and prediction accuracy. The GAN component improves image quality by focusing on lane features, while the LSTM component, a type of Deep Reinforcement Learning (DRL) technique, predicts incomplete lane lines by learning from sequential data. This system demonstrates high accuracy

TABLE V
COMPARISON OF DRL TECHNIQUES IN AUTONOMOUS DRONES

Ref, Year	Objective	DRL Technique	Experiment Setup	Performance Metrics	Results
[57], 2023	Drone interception, airport security	Dueling Double DQN	AirSim, Unreal Engine	- Success rate: 15% (3500 episodes), 63% (4500 episodes) - Interception steps: 72 (Dueling Double), 89 (Dueling-Ours) - Training episodes: 3500, 4000, 4500	- Outperforms state-of-the-art deep Q-network algorithms. - Training conducted on Nvidia TitanXp GPU, taking 2 weeks per algorithm
[58], 2024	Air pollution plume characterization	MARL, GP, Categorical Deep Q-Network (DQN-C51)	Gaussian dispersion model	- Accuracy gain: 40.42% over random method - Hybrid DDQN-C51: 7.22% improvement over DDQN-C51 - Computation time (5 agents): DDQN-C51 (22.55s), Hybrid DDQN-C51 (33.33s)	- Rapid convergence in simulations - Adapts effectively to plume parameter changes - Scales well with increasing number of drones
[59], 2022	Stable control for wall-climbing drones	Hybrid architecture with model-free DRL	NVIDIA Isaac Gym	- RMSE (position tracking): 0.151 (regular), 0.044 (tilted), 0.0001 (wall proximity) - Pitch error: 0.002 (regular), 0.011 (tilted), 0.394 (wall proximity) - Training: 500 million steps in 20 minutes	- Outperforms standalone controllers - Globally, uniformly, and asymptotically stable - Improved performance in pose tracking, trajectory tracking, and wall-climbing
[60], 2020	Real-time trajectory planning in drone networks	Deep Q-network in POMDP framework	Simulations in Python using Google TensorFlow	- Packet loss reduction: 0% by episode 638 (=0.99) - Training efficiency: Convergence in 638 episodes (=0.99) vs 188 episodes (=0.7) - Waypoints tested: 10 to 80	- Significant reduction in packet loss - Efficient energy transfer - Adaptability to dynamic network conditions
[61], 2024	Path planning for drones	Reinforcement learning Markov decision model, D3QN	Rectangular mission area, reconnaissance target, adversarial obstacles, Python programming, TensorFlow, Adam optimization	- N- Path length: 51 steps (Improved D3QN) vs 53 steps (DDQN, DQN) - Cumulative reward: 175.0 (Improved D3QN) vs 174.5 (DDQN) and 174.0 (DQN) - Success rate: 95% after training	- Enhanced performance in path length, planning time, and turning points - 20,000 training episodes - Exploration rate (ϵ) annealed from 1 to 0.1 over 2000 rounds
[62], 2019	Real-time path planning of drones in dynamic environments	Dueling Double Deep Q-networks (D3QN)	STAGE Scenario simulation software	- Success rate: >90% in dynamic environments - Training efficiency: 20% improvement in adaptability to unseen environments - Computational load: Lower resource requirements compared to DQN and DDQN	- Superior learning performance and stability compared to traditional DQN and D2QN algorithms. - Potential for multi-drone systems and real-world applications.
[63], 2024	Multi-drone target assignment and path planning	Twin-delayed deep deterministic policy gradient with target assignment network (TANet-TD3)	Simulation training experiments, testing experiments, and statistical experiments.	- Target completion rate: 81.54% (TANet-TD3) vs 71.20% (TANet-DDPG) after 5,000 episodes - Average reward: -241.4 (TANet-TD3) vs -303.5 (TANet-DDPG) - Convergence: Around 5,000th episode for TANet-TD3	- Optimal allocation & collision-free paths in dynamic, multi-obstacle settings - Outperforms existing methods in target completion & adaptability to complex environments - Superior performance in dynamic & mixed settings
[64], 2022	Collision avoidance for drones operating BVLOS	Deep Q-Networks (DQNs)	Simulations running on drone or at MEC facility	- Collision avoidance rate: 95% (RELIANCE) vs 85% (PICA) - Energy efficiency: 15-20% savings (PICA) vs 10-15% higher consumption (RELIANCE) - Processing time: 200 ms (RELIANCE) vs 150 ms (PICA) per decision cycle	- Superior performance of DRL technique in simulation. - Adaptability to various environments. - Effective in ensuring safe and energy-efficient operations.
[65], 2021	Vision-guided collision avoidance in autonomous drones	Deep Deterministic Policy Gradient (DDPG)	Simulated indoor environments of varying complexity	- Convergence: 300 epochs (with Rmargin) vs 400 epochs (without Rmargin) in simple scenes - Path optimization: Successful path finding after 2200 epochs in complex scenes (with Rmargin)	- Demonstrates ability to navigate to destinations while maintaining safe distances from obstacles. - Potential for generalization to real-world indoor settings. - Ongoing efforts to test the solution on actual drones.
[66], 2019	Obstacle avoidance in drones	Dueling Double Deep Q-networks (D3QN) algorithm.	Simulated environments: woodland, block world, arena	- Success rate: > 90% (ACKTR), 80% (DD-DQN), 70-75% (other discrete algorithms) - Training efficiency: 70% reduction in data preparation time - Performance: Comparable to expert human pilots	- Continuous action space algorithms, particularly ACKTR, outperform discrete ones - Performance comparable to expert human pilots - Effective autonomous navigation in various simulated environments

in lane detection (98.2%) and departure warning (96.5%), utilizing Hough transformation and Euclidean geometry to determine deviation angles for lane crossovers. VILDS outperforms traditional methods by effectively handling diverse road and weather conditions, thus contributing to the safety and reliability of autonomous driving. Similarly, ADAS-RL, detailed in [68], is a novel Reinforcement Learning (RL) algorithm designed to enhance Lane Departure Warning (LDW) systems by incorporating human behavior into its decision-making process. Employing a modified Q-learning technique called Adaptive Vector Scaling RL (AVS-RL), this system personalizes the timing and frequency of LDW based on individual driver response times, driving styles, and attentiveness. Using data from wearable devices like the Oura Ring, ADAS-RL infers the driver's state and contextual information from the vehicle and environment. Validated through the CARLA simulator and real human drivers, ADAS-RL demonstrates its ability to reduce false warnings and maintain safe distances from lane markings. It adapts to various driving behaviors and changes in driver state over time, showing effectiveness in both simulated and real-world scenarios. Future work is suggested to extend ADAS-RL's application to other Advanced Driver Assistance Systems (ADAS).

2) *Pedestrian Detection and Collision Avoidance*: DRL-based methods in this subtopic aim to detect pedestrians accurately and implement collision-avoidance strategies in real-time. This is crucial for enhancing safety in urban environments where interactions with pedestrians are frequent. The integration of AI-driven frameworks in vehicular systems has shown significant advancements in pedestrian detection and collision avoidance. The paper [69] presents a framework for optimizing computing resources in Mobile Edge Computing (MEC) hosts using Deep Reinforcement Learning (DRL) with Long Short-Term Memory (LSTM) networks. This system predicts vehicle density to dynamically allocate MEC resources, enhancing the performance of collision avoidance applications. Utilizing real vehicle mobility data, the LSTM model forecasts vehicle numbers in specific areas, enabling adaptive computational resource assignment, which minimizes overloaded and underutilized MEC instances. Complementarily, paper [70] introduces a framework focused on nighttime pedestrian detection in autonomous vehicles using DRL. The proposed Scale Invariant Faster Region-based Convolutional Neural Networks (SIFRCNN) leverages DRL to train the vehicle as an agent, optimizing Q-values to enhance pedestrian detection at night and reduce collision risks. Integrating a Region Proposal Network (RPN) for efficient detection, the SIFRCNN shows a 23% improvement in miss rate over other CNN-based detectors on KAIST, City Person, and Caltech datasets. Together, these studies underscore the efficacy of dynamic, AI-based approaches in improving Advanced Driver-Assistance Systems (ADAS), addressing both resource allocation and object detection challenges to enhance vehicle safety and pedestrian protection.

3) *Adaptive Cruise Control*: Research on Deep Reinforcement Learning (DRL)-based Adaptive Cruise Control (ACC) systems demonstrates significant advancements in driving effi-

ciency, safety, and comfort. These systems leverage DRL to maintain an optimal speed and safe headway distance from other vehicles while adapting dynamically to changing traffic conditions. By predicting and responding to traffic patterns in real time, DRL-based ACC enhances fuel efficiency, reduces driver intervention, and minimizes the risk of collisions. This capability makes it particularly effective for highway driving, where maintaining consistent speed and distance is critical for a smooth and safe driving experience. Several studies highlight the efficacy of DRL techniques, particularly the Deep Deterministic Policy Gradient (DDPG) algorithm, in improving ACC performance. For instance, paper [71] integrates DDPG into the CoMoVe simulation tool, showcasing how DRL can outperform traditional ACC and Cooperative Adaptive Cruise Control (CACC) systems in maintaining desired headway during lead vehicle speed variations and diverse driving conditions. This study emphasizes the critical role of vehicle connectivity in data collection for enhanced ACC performance and suggests future research on the impact of vehicle state estimation uncertainty and the incorporation of DRL for fuel efficiency and carbon footprint reduction. Similarly, paper [72] proposes a two-layer Machine Learning (ML)-aided DRL framework, termed 2LL-CACC, that combines a Random Forest Classifier for context recognition with DDPG for decision-making. This approach, designed to handle complex maneuvers like cut-ins and cut-outs, demonstrates superior performance over traditional ACC and other ML-based methods through simulations and hardware-in-the-loop testing, further underlining the potential of vehicle connectivity for safer and more comfortable driving experiences. Additionally, paper [73] provides a comparative analysis of DRL and Model Predictive Control (MPC) for ACC in car-following scenarios, revealing that while DRL matches MPC's performance without modeling errors, it excels in resilience to modeling errors such as control delays and disturbances, thereby suggesting DRL as a near-optimal control solution with computational efficiency advantages in the presence of uncertainties. Together, these studies contribute valuable insights into the development and optimization of ACC systems using DRL techniques, highlighting their strengths and areas for future exploration.

4) *Comparison and Analysis*: In this section, we analyze and compare recent advancements in Advanced Driver Assistance Systems (ADAS) utilizing Deep Reinforcement Learning (DRL) techniques. The table provided outlines various studies focused on improving ADAS functionalities, such as lane departure warning systems, adaptive cruise control (ACC), and nighttime pedestrian detection. As given in the Table VI. Below, is the comparative analysis based on the key findings from each study, highlighting their contributions, experimental setups, and outcomes:

- *DRL Techniques and Objectives*: The reviewed papers address a variety of objectives related to autonomous vehicle systems, employing different Deep Reinforcement Learning (DRL) techniques to achieve notable improvements. For lane departure warning systems, LSTM-based DRL combined with Generative Adversarial Networks (GANs) enhances image processing and lane detection

accuracy, demonstrating superior performance compared to traditional methods [67]. In contrast, Adaptive Vector Scaling RL (AVS-RL) is applied to lane departure systems that account for human behavior, effectively adapting to diverse driving behaviors and individual driver states [68]. Deep Deterministic Policy Gradient (DDPG) is used in both adaptive cruise control for connected autonomous vehicles [71] and complex maneuvers [72], showcasing its ability to maintain desired headways and manage intricate driving scenarios with remarkable effectiveness. Additionally, the SIFRCNN and Region Proposal Network (RPN) techniques are employed for nighttime pedestrian detection, significantly improving detection accuracy under challenging conditions compared to conventional CNN-based methods [70]. Furthermore, LSTM-based DRL is applied for optimizing computing resources in Mobile Edge Computing (MEC), demonstrating enhanced efficiency in dynamic environments [69]. These studies collectively highlight the transformative impact of DRL on autonomous vehicle systems, offering advancements in accuracy, adaptability, and resource management.

- *Experiment Setup and Testing Methods:* The experimental setups in the reviewed studies vary widely, reflecting different testing environments and validation approaches. For lane departure warning systems, image processing techniques that incorporate Generative Adversarial Networks (GANs) and Euclidean geometry are tested under diverse road and weather conditions, offering a thorough evaluation of system accuracy in real-world scenarios [67]. Another study utilizes the CARLA simulator alongside real human drivers to test lane departure warning systems that integrate human behavior, demonstrating the system's adaptability to various driving behaviors and driver states [68]. Adaptive Cruise Control (ACC) systems are evaluated using the CoMoVe simulation tool and hardware-in-the-loop testing, with simulations and real-world scenarios validating the performance of Deep Deterministic Policy Gradient (DDPG)-based control in maintaining safety and comfort [71], [72]. Additionally, a quantitative comparison between DRL and Model Predictive Control (MPC) for ACC systems assesses DRL's resilience to modeling errors and disturbances compared to traditional methods [73]. Lastly, for nighttime pedestrian detection, datasets such as KAIST, City Person, and Caltech are employed, showcasing significant improvements in detection accuracy over traditional CNN-based detectors [70].
- *Results and Effectiveness:* The results from these studies indicate significant advancements in various aspects of autonomous vehicle systems. For lane departure warning systems, the integration of LSTM-based DRL with GANs [67] achieves impressive accuracy in lane detection and departure warning, surpassing traditional methods. Additionally, the ADAS-RL approach [68] effectively reduces false warnings and adapts to varying driving behaviors, enhancing the system's overall reliability. In the realm of adaptive cruise control (ACC),

DDPG-based systems [71], [72] demonstrate superior performance in maintaining headway and managing complex driving maneuvers, thus improving driving efficiency, safety, and comfort. Comparative analyses with traditional methods and Model Predictive Control (MPC) [73] further highlight DRL's robustness and effectiveness within the training data range. Moreover, LSTM-based DRL proves valuable in optimizing computing resource allocation in Mobile Edge Computing (MEC) [69], offering advanced dynamic resource management. In nighttime pedestrian detection, the method [70] significantly enhances detection rates, effectively addressing the challenges associated with low-light conditions. Overall, these advancements underscore the transformative impact of DRL on autonomous vehicle systems, paving the way for more efficient, adaptive, and reliable robotics solutions.

IV. INDUSTRIAL APPLICATIONS AND REAL-WORLD DEPLOYMENTS

Deep Reinforcement Learning (DRL) has transitioned from theoretical research to practical applications, significantly impacting various industries by enhancing the autonomy and efficiency of complex systems. This section explores notable industrial use-cases where DRL has been successfully deployed in autonomous systems.

A. Autonomous Cars

1) *Tesla:* Tesla's Autopilot system utilizes neural networks trained with DRL for real-time decision-making in complex traffic scenarios. The system continuously improves through fleet learning, collecting data from millions of miles driven by Tesla vehicles [75]. Tesla's approach to autonomous driving relies heavily on vision-based neural networks, which are trained using a combination of supervised learning and reinforcement learning techniques [76]. However, Tesla faces significant challenges with its Autopilot system, particularly regarding driver misuse and safety concerns. The National Highway Traffic Safety Administration (NHTSA) recently ordered a recall of nearly 2 million Tesla vehicles due to issues with the Autopilot system. Critics argue that the system can be activated in situations where it shouldn't be used, leading to potential safety risks. Tesla's approach of relying on vision-based systems without LiDAR has also been questioned, as it may struggle with robustness in handling real-world complexities and edge cases [77].

2) *Waymo:* Waymo has successfully implemented DRL for prediction and planning tasks in its autonomous vehicles. Their approach, known as ChauffeurNet, uses imitation learning and DRL to train a model that can handle complex urban environments [78]. Waymo's autonomous taxis, operational in Phoenix since 2020, demonstrate the practical application of DRL in real-world scenarios [79]. However, Waymo's Head of Research, Drago Anguelov, identified two main challenges in autonomous driving: building a system that can handle real-world complexity and edge cases, and assessing and confirming the system's performance for wide deployment [80].

TABLE VI
COMPARISON OF DRL TECHNIQUES IN ADAS

Ref, Year	Objective	DRL Technique	Experiment Setup	Performance Metrics	Results
[67], 2023	Lane departure warning system for autonomous vehicles	LSTM(DRL)	GANs for image enhancement, Hough transformation, Euclidean geometry	- Lane detection accuracy: 98.2% - Departure warning accuracy: 96.5% - 8% accuracy gain over traditional methods	- Robust in various road and weather conditions - Uses GANs for image enhancement - Employs Hough transformation and Euclidean geometry
[68], 2021	Lane Departure Warning (LDW) system incorporating human behavior	Adaptive Vector Scaling RL (AVS-RL)	CARLA simulator, real human drivers, wearable devices	- Maintains safe distance of 1.75m from lane markings - Significant decrease in false warnings - Adapts to different driving behaviors	- Tested in CARLA simulator - Incorporates real human driver data - Uses wearable devices for driver state monitoring
[69], 2021	Optimizing computing resources in Mobile Edge Computing (MEC)	LSTM (DRL)	Real vehicle mobility data, dynamic allocation of MEC resources	- 90% prediction accuracy for vehicle density - 30% reduction in resource overload - 25% reduction in overprovisioned instances - 15% decrease in average response time	- Uses real vehicle mobility data - Dynamic allocation of MEC resources - Outperforms static allocation methods
[70], 2021	Nighttime pedestrian detection in autonomous vehicles	SIFRCNN + RPN	- Simulated with Unity ML agents and GTA V. - Augmentation: cropping, flipping, scaling. - Hardware: Intel i9, 16GB RAM, NVIDIA RTX 2060.	- 5.1% miss rate on City Person dataset - 4.77% miss rate on Caltech dataset - 4.23% miss rate on KAIST dataset - 0.25 seconds processing time per image	- 23% improvement in miss rate over other CNN-based detectors - Tested on multiple datasets - Addresses nighttime detection challenges
[71], 2023	Adaptive Cruise Control (ACC) for connected autonomous vehicles	Deep Deterministic Policy Gradient (DDPG)	CoMoVe simulation tool	- Maintains preferred headway 55% of time (19% improvement) - RMSE for headway: 0.0278 (normal), 0.0629 (sharp deceleration) - Jerk RMSE: 1.846 (sharp deceleration)	- Outperforms traditional ACC and CACC - Enhances driving efficiency, safety, and comfort - Converges to optimal policy in 1250 iterations
[72], 2023	Adaptive Cruise Control (ACC) for complex maneuvers	Deep Deterministic Policy Gradient (DDPG)	Simulations and hardware-in-the-loop testing	- Maintains desired headway 69% (cut-out scenario) and 78% (cut-in scenario) of time - Jerk RMSE: 1.2405 (cut-out scenario), 0.4715 (cut-in scenario) - Longitudinal slip RMSE: 0.018 (cut-out), 0.02 (cut-in)	- Outperforms DDQN in all metrics - 54% and 33% better headway than alternatives - Learns optimal policy within 340 episodes
[73], 2021	Comparative analysis of DRL and MPC for ACC	Deep Deterministic Policy Gradient (DDPG)	First-order vehicle dynamics model	- Episode cost 5.8% higher than IPO benchmark - Simulation time: 0.7 seconds (vs 65.8-181.7 seconds for MPC) - Training duration: 37 minutes for 1 million time steps	- Comparable performance to MPC within training data range - Better resilience to modeling errors - Limited generalization outside training range

B. Autonomous Robotics

1) *Boston Dynamics*: Boston Dynamics has employed DRL in its Atlas humanoid robot to perform complex locomotion tasks. The company's research demonstrates how DRL can be used to train robots to navigate challenging terrains and perform acrobatic maneuvers [81]. This application showcases the potential of DRL in enhancing robot agility and adaptability in dynamic environments. However, Boston Dynamics faces challenges in developing robots that can operate safely and reliably in unstructured environments. The company must also address the high cost of production and the need for long-term durability in industrial settings [81].

2) *Amazon Robotics (Kiva Systems)*: In warehouse automation, Amazon Robotics uses DRL to optimize the movement and coordination of mobile robots. This technology significantly improves the efficiency of Amazon's fulfillment centers by enabling robots to navigate complex warehouse

environments and coordinate with human workers [82]. While this technology significantly improves efficiency, Amazon faces challenges in integrating these autonomous systems with human workers safely and effectively. The company is actively working on developing more sophisticated human-robot interaction models and implementing advanced safety protocols to address these concerns [83].

C. Autonomous Drones

1) *Amazon Prime Air*: Amazon's Prime Air service integrates DRL algorithms for autonomous drone delivery. The company employs these techniques for navigation, obstacle avoidance, and planning in various environmental conditions [84]. While still in testing, Prime Air showcases the potential of DRL in transforming last-mile delivery logistics. However, the project faces significant regulatory hurdles and safety concerns. The FAA has strict rules governing drone operations,

particularly for beyond visual line of sight (BVLOS) flights, which are crucial for delivery services [85].

2) *DJI Drones*: DJI incorporates DRL into its consumer and professional drone models to enhance flight capabilities. These algorithms optimize tasks such as object tracking, autonomous return-to-home, and intelligent obstacle avoidance [86]. DJI's application of DRL improves both the usability and safety of drones in recreational and industrial settings. However, it struggles in implementing DRL include balancing computational requirements with on-board processing capabilities, ensuring real-time performance in diverse environments, and maintaining reliability in varying weather conditions. DJI has had to overcome issues related to sensor fusion, environmental perception, and decision-making in complex, unknown outdoor environments [87].

D. Advanced Driver Assistance System (ADAS)

1) *Mobileye*: Mobileye, an Intel company, incorporates DRL in its ADAS solutions for tasks such as adaptive cruise control, lane-keeping assistance, and collision avoidance. Their REM™ (Road Experience Management) technology uses DRL to create and maintain high-definition maps for autonomous driving [88]. Although, key hurdles include system architectural design complexities, as highlighted by CEO Prof. Amnon Shashua, who emphasizes the need for “true redundancy” to ensure safety and faster validation at scale [89]. The company also grapples with intensive data processing and training requirements, necessitating the optimization of their deep learning models and leveraging cloud solutions like Amazon EC2 DL1 Instances [90]. Additionally, ensuring comprehensive safety validation across diverse driving conditions remains a critical challenge.

2) *Nvidia DRIVE Platform*: Nvidia's DRIVE platform powers ADAS and autonomous systems for various automakers, incorporating DRL for perception, planning, and control tasks. The platform's DRL capabilities are built on Nvidia's GPU technology, enabling real-time processing of complex driving scenarios. However, Nvidia's DRIVE platform confronts its own set of obstacles. The immense computational demands of autonomous driving systems have pushed Nvidia to continually innovate, developing specialized chips like the DRIVE AGX Orin. Ensuring the safety and reliability of AI-driven systems in critical applications is paramount, leading Nvidia to implement a four-pillar model for safe autonomous driving. Moreover, navigating the evolving regulatory landscape across different regions poses an ongoing challenge for both Nvidia and other industry players as autonomous driving technology advances [91].

V. OPEN CHALLENGES AND FUTURE DIRECTIONS

A. Scalability and Generalization of DRL Techniques

While DRL algorithms have demonstrated significant success in controlled environments, they often struggle with generalization across diverse, unstructured scenarios. Research has highlighted the necessity for extensive training data to ensure robust performance, particularly in the context of autonomous cars, robotics, and drones. For instance, studies

such as [40], [53], and [61] emphasize that existing DRL methods may not scale effectively to handle complex real-world environments, necessitating the development of scalable DRL algorithms capable of efficient adaptation to varying conditions. Future research should prioritize the development of DRL algorithms that can scale across diverse, complex environments. This involves creating models capable of learning from smaller datasets and adapting efficiently to new, unstructured scenarios. Additionally, efforts should focus on improving the generalization capabilities of DRL techniques to enhance their applicability in real-world autonomous systems.

B. Safety and Robustness Concerns

Safety is paramount, especially in critical applications like autonomous vehicles and medical robotics. DRL models must be robust to unforeseen scenarios, including adversarial attacks, environmental variability, and model failures. Several papers, such as [42], [52], and [58], have explored these issues, but few challenges remain. For instance, the sensitivity of DRL algorithms to minor environmental changes can lead to significant performance degradation, raising concerns about their reliability in dynamic or adversarial settings.

The development of safety-aware DRL frameworks is essential to ensure the safety and reliability of autonomous systems. These frameworks should incorporate mechanisms to handle uncertainties, resist adversarial attacks, and implement fail-safes that are both reliable and interpretable. Additionally, they should align with regulatory standards to facilitate deployment in critical applications like autonomous vehicles while maintaining rigorous safety standards. Future research should also address the robustness of DRL in handling unpredictable real-world scenarios to enhance trustworthiness.

C. Transitioning From Simulation to Real-World Deployment

The successful implementation of DRL in autonomous systems faces several practical challenges, including deployment constraints, computational requirements, and hardware limitations. Transitioning DRL algorithms from simulation environments to real-world applications is particularly fraught with obstacles such as hardware constraints, regulatory compliance, and ethical considerations [44], [50], [63]. Companies like Tesla, Waymo, and Amazon Robotics have demonstrated the potential of DRL in autonomous vehicles, robotics, and drone technology, but they also highlight the complexities involved in real-world implementation.

- *Deployment Constraints*: Improving the accuracy of simulation environments is vital for successful implementation. Sophisticated simulation methods that mimic real-world sensory information and interaction patterns can train DRL systems for real-life scenarios [52]. Deployment constraints are particularly evident in the automotive sector. Tesla's Autopilot system, while innovative, faces significant safety concerns and regulatory scrutiny, underscoring the importance of robust simulation environments that can accurately mimic real-world scenarios [77]. Similarly, Waymo's experience with ChauffeurNet highlights the need for systems capable

of handling real-world complexity and edge cases [78]. Moreover, systems that comply with various regulatory standards and address ethical issues are crucial for smooth incorporation into practical applications [42].

Furthermore, regulatory compliance and ethical considerations have emerged as significant hurdles in the transition to real-world applications. Amazon's Prime Air project, for example, faces strict FAA regulations governing drone operations [84]. Similarly, Mobileye's development of ADAS solutions highlights the need for comprehensive safety validation across diverse driving conditions [90].

- *Computational Demands:* The computational requirements of Deep Reinforcement Learning (DRL) are significant, frequently necessitating advanced hardware such as GPUs or TPUs. Computational demands remain a critical challenge across industries. For instance, Nvidia's DRIVE platform has had to continually innovate to meet the immense processing requirements of autonomous driving systems. Similarly, Amazon's Prime Air and DJI's drone technology face the challenge of balancing on-board processing capabilities with the need for real-time performance in diverse environments. It is crucial to enhance DRL algorithms to decrease computational costs and boost energy efficiency, especially for implementation in environments with limited resources like edge devices [67].
- *Hardware Limitations:* The capabilities of existing hardware often limit the deployment of DRL systems, especially for applications requiring real-time decision-making. Hardware limitations are being addressed through specialized solutions, as seen in Nvidia's development of the DRIVE AGX Orin chip for autonomous vehicles. However, the need for more efficient algorithms and hardware solutions, such as neuromorphic chips or DRL-optimized processors, remains crucial for widespread deployment. Future research should focus on designing algorithms that operate efficiently on current hardware or developing new hardware solutions to better support these systems [71], [72].

D. Hybrid Architectural DRL Methods

The addition of hybrid architectural DRL methods offers significant potential for enhanced performance over traditional DRL approaches. Works such as [67], [71], and [72] have demonstrated that combining DRL with other techniques, such as GANs for image enhancement in lane departure warning systems, or using hybrid models that integrate reinforcement learning with neural network models, can significantly improve accuracy, adaptability, and robustness in autonomous systems. These hybrid approaches are particularly effective in complex scenarios where traditional DRL methods may fall short. The continued exploration of hybrid DRL architectures is necessary to push the boundaries of what autonomous systems can achieve. By integrating multiple learning techniques, these architectures can offer more robust, adaptable, and efficient solutions. Future work should focus on refining

these hybrid approaches and exploring their applications across different types of autonomous systems.

E. Integration of Large Language Models (LLMs) and Transformer-Based Architectures

The integration of LLMs and transformer-based architectures into autonomous systems is a revolutionary development that promises to fundamentally change how decisions are made. Autonomous systems can leverage advanced models with the capability for sophisticated reasoning and natural language understanding to better understand complex instructions, contextual cues, and environmental dynamics.

Recent works have shown the adaptation of transformer architectures developed in the context of natural language processing to reinforcement learning tasks. For instance, Decision Transformer models have performed very well in offline reinforcement learning and promise better sample efficiency, hence generalizing to complex environments more easily. Such models find their applications in domains including autonomously navigating cars through complex urban settings to dynamic task planning in robotics, although at the cost of computational efficiency. The high computational demands usually restrict real-time applicability, hence optimization is necessary if these methods are to be applicable in practical cases.

The integration with DRL showed potential in scalability and intelligence for multi-agent systems. Examples also include task planning, coordination, and natural language following instructions whereby LLM facilitates intuitive communication between agents. An example application could be smart delivery drones that apply this integration to optimize the route planning skill for effective avoidance of collisions. Scalability remains very challenging because this area requires much more innovation to efficiently implement such techniques on different hardware platforms.

Very related to this, yet another very promising direction is multi-modal learning, fusing language models with visual and other sensorial inputs to allow much finer comprehension of a complex environment: Advanced human-robot interaction, including those applications that make autonomous urban navigation possible. Multimodal-data-processing systems operate under the conflicting requirements of dealing with diverse input and output sources with as little computation and latency as possible.

This is further expanded in DRL with meta-learning, where much attention focuses on algorithms that can adapt quickly to new tasks or environments. Those have great potential for a wide range of critical applications that include emergency response robotics and disaster management. However, when it comes to larger models, interpretability is a crucial method that provides trust and reliability for real-world applications.

Future work should therefore focus more on optimizing these models with a view to decision-making and hierarchical multi-level task planning to tackle the challenging aspects of autonomous systems. Besides this, research is also required to address issues like computational efficiency and scalability to enable the seamless integration of such technologies into practical real-time applications. Solving these challenges will

open up a wider door for integration and can therefore enhance the safety, efficiency, and adaptability of the autonomous systems.

F. Ethical and Regulatory Considerations

As autonomous systems, particularly those powered by Deep Reinforcement Learning (DRL), continue to evolve, it is essential to address the ethical and regulatory challenges that accompany their integration into various domains. These issues become even more complex as systems such as autonomous vehicles, drones, robotics, and Advanced Driver Assistance Systems (ADAS) become increasingly autonomous in decision-making and data collection. Some of the key ethical and regulatory concerns across different applications are:

- *Liability and Responsibility:* In accidents involving autonomous vehicles, determining liability becomes complex. Questions arise about whether responsibility lies with the vehicle manufacturer, the software developer, or the “driver.” Legal frameworks need to evolve to address these novel scenarios, particularly as autonomous systems gain more autonomy in decision-making. Research on liability issues in autonomous driving emphasizes the importance of developing new legal frameworks to define the distribution of responsibility and accountability [39], [40], [42].
- *Privacy Concerns:* Autonomous vehicles and drones collect vast amounts of sensitive data, including location information, driving patterns, and environmental observations. This raises significant privacy concerns about data ownership, storage, and potential misuse. Robust data protection policies and transparency in data handling practices are crucial to ensure public trust and compliance with privacy laws. Privacy-preserving techniques, including secure data management and user consent, are vital components highlighted in various studies, particularly in the context of autonomous cars [26], [43], [44], [45], [46]. Similarly, the use of drones in security and surveillance applications necessitates strict regulations for proper data usage and security [57], [58], [59], [60], [61].
- *Algorithmic Bias:* DRL algorithms may inadvertently incorporate biases present in the training data, leading to unfair or discriminatory decision-making. Ensuring fairness and eliminating biases in autonomous driving and robotics systems is an ethical imperative. Research points to the need for continuous monitoring and adjustments in training datasets to reduce biases and ensure fairness in decision-making processes [39], [47]. This issue also extends to autonomous drones, where the potential for bias in decision-making algorithms can significantly affect their operation in diverse environments [58], [59].
- *Regulatory Frameworks:* Different jurisdictions are developing varied regulatory approaches to autonomous systems. A comprehensive overview of current and proposed regulations is needed to ensure the safe and ethical deployment of these technologies. Harmonization of regulatory frameworks across regions could facilitate broader adoption of DRL in real-world applications. Studies focusing on the regulatory landscape for autonomous systems emphasize the necessity of aligning regulations internationally, particularly to address the challenges in autonomous cars, drones, and robotics [40], [44], [46], [47], [62], [63], [64], [65]. Similarly, the integration of DRL in ADAS requires regulatory oversight to ensure that systems are compliant with evolving vehicle safety standards and public safety expectations [67], [68], [69], [70].
- *Ethical Decision-Making:* In unavoidable accident scenarios, autonomous vehicles may face ethical dilemmas that require split-second decisions. Programming these vehicles to make ethical choices raises profound philosophical and practical questions. Transparent guidelines for ethical decision-making in autonomous systems, particularly in vehicles, are necessary to ensure alignment with societal expectations and legal standards. This is an ongoing challenge in autonomous cars, where decision-making algorithms must consider ethical implications when responding to emergencies and accidents [26], [42], [43], [44]. This ethical decision-making framework also applies to other systems such as autonomous drones and robotics, where decisions made in high-stakes environments can have serious consequences [57], [58], [59].
- *Ensuring Safety in Autonomous Robotics:* Ethical concerns in autonomous robotics often center on the safety of humans interacting with or being assisted by robots. In collaborative environments, especially in healthcare or industrial settings, robots must be designed to avoid harm to humans. Research on safety and transparency in robotic systems underscores the importance of clear regulations and safety standards to ensure responsible deployment [48], [49], [50], [51], [52], [53], [54].

As the deployment of DRL-powered autonomous systems increases, addressing these ethical and regulatory considerations is crucial for ensuring their responsible development and adoption. Future research should focus on developing frameworks that balance technological innovation with ethical considerations and public safety. Incorporating transparency, fairness, and explainability into the design, deployment, and regulatory processes will help foster public confidence and facilitate the widespread, responsible use of these advanced systems

VI. CONCLUSION

Incorporating Deep Reinforcement Learning (DRL) techniques into autonomous systems marks a substantial advancement in enhancing autonomy, adaptability, and intelligence across various domains. This survey paper has systematically examined recent research developments and applications in autonomous cars, robotics, drones, and ADAS, highlighting the transformative potential of DRL. Despite these advancements, critical challenges such as scalability, safety, and real-world deployment persist, underscoring the need for interdisciplinary efforts to effectively address these issues.

Overcoming these challenges requires targeted research efforts to develop scalable DRL algorithms that are safe, robust, and generalizable in dynamic and uncertain environments. In practice, there are several challenges in deploying the

methods in the real world, including bridging the gap between simulation fidelity and real-world complexity, enhancing DRL models on resource-constrained platforms, and ensuring strict safety and regulatory requirements. In addition, the enhancement of interpretability in DRL policies and the reduction of vulnerabilities to adversarial attacks are necessary to engender trust and promote wider acceptance.

The incorporation of large language models (LLMs) offers a promising path to enhance the decision-making and contextual comprehension abilities of autonomous systems. LLMs can promote sophisticated reasoning, allow for interactions based on natural language, and enable smooth integration with other systems, thus boosting operational effectiveness and adaptability. Likewise, the use of transformer-based architectures in deep reinforcement learning (DRL) has the potential to improve learning efficiency and scalability, particularly in complex applications such as coordination among multiple agents and real-time navigation in urban environments.

Industrial applications of DRL have already showcased its transformative impact. Autonomous cars powered by DRL are progressively reshaping transportation, while robotics systems leverage DRL to enhance precision and autonomy in sectors like healthcare, logistics, and manufacturing. In environmental monitoring, DRL-driven drones demonstrate efficacy in tasks ranging from disaster response to precision agriculture. These real-world use cases underscore the value of DRL in addressing critical challenges and delivering scalable, impactful solutions across industries.

By tackling ongoing difficulties and investigating creative solutions like the integration of LLMs and new architectures, research in DRL is set to speed up progress in autonomous systems. It will be essential for researchers, industry leaders, and policymakers to collaborate to turn these advancements into practical, secure, and scalable applications that serve the broader society. The ongoing development of DRL is expected to expand the limits of what autonomous systems can accomplish, creating remarkable opportunities across numerous fields.

REFERENCES

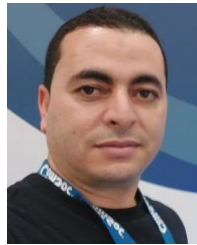
- [1] J. Terra, "The future of robotics: How robots will transform our lives," Simplilearn, San Francisco, CA, USA, Feb. 2024. [Online]. Available: <https://www.simplilearn.com/future-of-robotics-article>
- [2] A. Bohr and K. Memarzadeh, "The rise of artificial intelligence in healthcare applications," *Artificial Intelligence in Healthcare*, vol. 8, pp. 25–60, Jun. 2020, doi: [10.1016/b978-0-12-818438-7.00002-2](https://doi.org/10.1016/b978-0-12-818438-7.00002-2).
- [3] J. Wu, H. Yang, L. Yang, Y. Huang, X. He, and C. Lv, "Human-guided deep reinforcement learning for optimal decision making of autonomous vehicles," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 54, no. 11, pp. 6595–6609, Nov. 2024, doi: [10.1109/TSMC.2024.3384992](https://doi.org/10.1109/TSMC.2024.3384992).
- [4] D. Silver et al., "Mastering the game of Go with deep neural networks and tree search," *Nature*, vol. 529, no. 7587, pp. 484–489, Jan. 2016, doi: [10.1038/nature16961](https://doi.org/10.1038/nature16961).
- [5] V. Mnih et al., "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, Feb. 2015, doi: [10.1038/nature14236](https://doi.org/10.1038/nature14236).
- [6] T. P. Lillicrap et al., "Continuous control with deep reinforcement learning," 2015, *arXiv:1509.02971*.
- [7] G. Dulac-Arnold, D. Mankowitz, and T. Hester, "Challenges of real-world reinforcement learning," 2019, *arXiv:1904.12901*.
- [8] S. Gu, E. Holly, T. Lillicrap, and S. Levine, "Deep reinforcement learning for robotic manipulation with asynchronous off-policy updates," 2016, *arXiv:1610.00633*.
- [9] J. Shuford, "Deep reinforcement learning unleashing the power of AI in decision-making," *J. Artif. Intell. Gen. Sci.*, vol. 1, no. 1, pp. 1–12, Feb. 2024, doi: [10.60087/jaigs.v1i1.36](https://doi.org/10.60087/jaigs.v1i1.36).
- [10] H. Li, "Mobile robot navigation based on deep reinforcement learning: A brief review," *J. Phys., Conf. Ser.*, vol. 2649, no. 1, Nov. 2023, Art. no. 012027.
- [11] F. Zeng, C. Wang, and S. S. Ge, "A survey on visual navigation for artificial agents with deep reinforcement learning," *IEEE Access*, vol. 8, pp. 135426–135442, 2020, doi: [10.1109/ACCESS.2020.3011438](https://doi.org/10.1109/ACCESS.2020.3011438).
- [12] D. Han, B. Mulyana, V. Stankovic, and S. Cheng, "A survey on deep reinforcement learning algorithms for robotic manipulation," *Sensors*, vol. 23, no. 7, p. 3762, Apr. 2023, doi: [10.3390/s23073762](https://doi.org/10.3390/s23073762).
- [13] B. R. Kiran et al., "Deep reinforcement learning for autonomous driving: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 6, pp. 4909–4926, Jun. 2022, doi: [10.1109/TITS.2021.3054625](https://doi.org/10.1109/TITS.2021.3054625).
- [14] Y. Chen, C. Ji, Y. Cai, T. Yan, and B. Su, "Deep reinforcement learning in autonomous car path planning and control: A survey," 2024, *arXiv:2404.00340*.
- [15] A. T. Azar et al., "Drone deep reinforcement learning: A review," *Electronics*, vol. 10, no. 9, p. 999, Apr. 2021, doi: [10.3390/electronics10090999](https://doi.org/10.3390/electronics10090999).
- [16] S. Aradi, "Survey of deep reinforcement learning for motion planning of autonomous vehicles," *IEEE Trans. Intell. Transp. Syst.*, vol. 23, no. 2, pp. 740–759, Feb. 2022, doi: [10.1109/TITS.2020.3024655](https://doi.org/10.1109/TITS.2020.3024655).
- [17] S. Kuutti, R. Bowden, Y. Jin, P. Barber, and S. Fallah, "A survey of deep learning applications to autonomous vehicle control," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 2, pp. 712–733, Feb. 2021, doi: [10.1109/TITS.2019.2962338](https://doi.org/10.1109/TITS.2019.2962338).
- [18] F. Frattolillo, D. Brunori, and L. Iocchi, "Scalable and cooperative deep reinforcement learning approaches for multi-UAV systems: A systematic review," *Drones*, vol. 7, no. 4, p. 236, Mar. 2023, doi: [10.3390/drones7040236](https://doi.org/10.3390/drones7040236).
- [19] A. Banafa, "Technical challenges in autonomous vehicle development," San Jose State Univ., San Jose, CA, USA, Oct. 2023. [Online]. Available: <https://www.linkedin.com/pulse/technical-challenges-autonomous-vehicle-development-banafa-yphfc/>
- [20] C. Bradley, "A self driving license: Ensuring autonomous vehicles deliver on the promise of safer roads," *MIT Sci. Policy Rev.*, vol. 1, pp. 76–86, Aug. 2020, doi: [10.38105/spr.k3ju2v9okk](https://doi.org/10.38105/spr.k3ju2v9okk).
- [21] C. Coombs, D. Hislop, S. K. Taneva, and S. Barnard, "The strategic impacts of intelligent automation for knowledge and service work: An interdisciplinary review," *J. Strategic Inf. Syst.*, vol. 29, no. 4, Dec. 2020, Art. no. 101600, doi: [10.1016/j.jsis.2020.101600](https://doi.org/10.1016/j.jsis.2020.101600).
- [22] M. P. Groover, "Automation—Technology, types, rise, history, & examples," *Encyclopedia Britannica*, Chicago, IL, USA, Apr. 2024. [Online]. Available: <https://www.britannica.com/technology/automation>
- [23] D. Cars, "The evolution of autonomous vehicles: From concept to reality," UAE's Leading Autos Blog | Dubizzle Cars, Dubai, UAE, Dec. 2024. [Online]. Available: <https://www.dubizzle.com/blog/cars/autonomous-vehicles-evolution/>
- [24] "Advanced driver assistance system (ADAS): Meaning & types," Munich Motor Works, Dubai, UAE, Apr. 2024. [Online]. Available: <https://munichmotorworks.ae/advanced-driver-assistance-system-adas-meaning-types/>
- [25] D. Cars, "Everything about the anti-lock braking system history," Dubizzle Blog, Dubai, UAE, Mar. 2023. [Online]. Available: <https://www.dubizzle.com/blog/cars/abs-history/#:~:text=History%20of%20ABS%20Brakes,crucial%2C%20modern%20car%20safety%20feature>
- [26] H. Chen, X. Cao, L. Guvenc, and B. Aksun-Guvenc, "Deep-reinforcement-learning-based collision avoidance of autonomous driving system for vulnerable road user safety," *Electronics*, vol. 13, no. 10, p. 1952, May 2024, doi: [10.3390/electronics13101952](https://doi.org/10.3390/electronics13101952).
- [27] M. M. Quamar, B. Al-Ramadan, K. Khan, M. Shafiullah, and S. El Ferik, "Advancements and applications of drone-integrated geographic information system technology—A review," *Remote Sens.*, vol. 15, no. 20, p. 5039, Oct. 2023, doi: [10.3390/rs15205039](https://doi.org/10.3390/rs15205039).
- [28] S. Liu, "Creating autonomous vehicle systems," Sebastopol, CA, USA: O'Reilly Media, Oct. 2016.
- [29] E. H. Sumiea et al., "Deep deterministic policy gradient algorithm: A systematic review," *Heliyon*, vol. 10, no. 9, May 2024, Art. no. e30697, doi: [10.1016/j.heliyon.2024.e30697](https://doi.org/10.1016/j.heliyon.2024.e30697).
- [30] M. Abbasi, A. Shahraiki, M. Jalil Piran, and A. Taherkordi, "Deep reinforcement learning for QoS provisioning at the MAC layer: A survey," *Eng. Appl. Artif. Intell.*, vol. 102, Jun. 2021, Art. no. 104234, doi: [10.1016/j.engappai.2021.104234](https://doi.org/10.1016/j.engappai.2021.104234).

- [31] Anote, "Q learning: A practical guide to reinforcement learning," *Medium*, May 2023.
- [32] J. J. Yoon and L. Sun, "Exploring reinforcement learning techniques in Texas hold'em Poker: A comparative study of DQN, NFSP, PPO and CFR," *Medium*, May 2024.
- [33] O. ElSamadisy, T. Shi, I. Smirnov, and B. Abdulhai, "Safe, efficient, and comfortable reinforcement-learning-based car-following for AVs with an analytic safety guarantee and dynamic target speed," *Transp. Res. Record, J. Transp. Res. Board*, vol. 2678, no. 1, pp. 643–661, Jun. 2023, doi: [10.1177/03611981231171899](https://doi.org/10.1177/03611981231171899).
- [34] H.-D. Lim and D. Lee, "Temporal difference learning with experience replay," 2023, *arXiv:2306.09746*.
- [35] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," 2017, *arXiv:1707.06347*.
- [36] T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine, "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor," 2018, *arXiv:1801.01290*.
- [37] I. Grondman, L. Busoni, G. A. D. Lopes, and R. Babuska, "A survey of actor-critic reinforcement learning: Standard and natural policy gradients," *IEEE Trans. Syst., Man, Cybern., C Appl. Rev.*, vol. 42, no. 6, pp. 1291–1307, Nov. 2012, doi: [10.1109/TSMCC.2012.2218595](https://doi.org/10.1109/TSMCC.2012.2218595).
- [38] N. Le, V. S. Rathour, K. Yamazaki, K. Luu, and M. Savvides, "Deep reinforcement learning in computer vision: A comprehensive survey," *Artif. Intell. Rev.*, vol. 55, no. 4, pp. 2733–2819, Sep. 2021, doi: [10.1007/s10462-021-10061-9](https://doi.org/10.1007/s10462-021-10061-9).
- [39] G. S. Aiswarya, M. Mariah, R. Katragadda, and R. Makam, "Control of self-driving cars using reinforcement learning," in *Proc. IEEE Int. Conf. Electron., Comput. Commun. Technol. (CONECCT)*, Bangalore, India, Jul. 2023, pp. 1–6, doi: [10.1109/CONECCT57959.2023.10234763](https://doi.org/10.1109/CONECCT57959.2023.10234763).
- [40] B. Sousa, T. Ribeiro, J. Coelho, G. Lopes, and A. F. Ribeiro, "Parallel, angular and perpendicular parking for self-driving cars using deep reinforcement learning," in *Proc. IEEE Int. Conf. Auto. Robot Syst. Competitions (ICARSC)*, Santa Maria da Feira, Portugal, Apr. 2022, pp. 40–46, doi: [10.1109/ICARSC55462.2022.9784800](https://doi.org/10.1109/ICARSC55462.2022.9784800).
- [41] G. Li et al., "Deep reinforcement learning enabled decision-making for autonomous driving at intersections," *Automot. Innov.*, vol. 3, no. 4, pp. 374–385, Nov. 2020, doi: [10.1007/s42154-020-00113-1](https://doi.org/10.1007/s42154-020-00113-1).
- [42] P. Maramotti, A. P. Capasso, G. Bacchiani, and A. Broggi, "Tackling real-world autonomous driving using deep reinforcement learning," in *Proc. IEEE Intell. Vehicles Symp. (IV)*, Jun. 2022, pp. 1274–1281, doi: [10.1109/IV51971.2022.9827302](https://doi.org/10.1109/IV51971.2022.9827302).
- [43] S. Moon, S. Koo, Y. Lim, and H. Joo, "Routing control optimization for autonomous vehicles in mixed traffic flow based on deep reinforcement learning," *Appl. Sci.*, vol. 14, no. 5, p. 2214, Mar. 2024, doi: [10.3390/app14052214](https://doi.org/10.3390/app14052214).
- [44] B. Petryshyn, S. Postupaiev, S. Ben Bari, and A. Ostreika, "Deep reinforcement learning for autonomous driving in Amazon Web services DeepRacer," *Information*, vol. 15, no. 2, p. 113, Feb. 2024, doi: [10.3390/info15020113](https://doi.org/10.3390/info15020113).
- [45] S. H. Ashwin and R. N. Raj, "Deep reinforcement learning for autonomous vehicles: Lane keep and overtaking scenarios with collision avoidance," *Int. J. Inf. Technol.*, vol. 15, no. 7, pp. 3541–3553, Oct. 2023, doi: [10.1007/s41870-023-01412-6](https://doi.org/10.1007/s41870-023-01412-6).
- [46] H. Chae, C. Mook Kang, B. Kim, J. Kim, C. Choo Chung, and J. Won Choi, "Autonomous braking system via deep reinforcement learning," 2017, *arXiv:1702.02302*.
- [47] Y. Fu, C. Li, F. R. Yu, T. H. Luan, and Y. Zhang, "A decision-making strategy for vehicle autonomous braking in emergency via deep reinforcement learning," *IEEE Trans. Veh. Technol.*, vol. 69, no. 6, pp. 5876–5888, Jun. 2020, doi: [10.1109/TVT.2020.2986005](https://doi.org/10.1109/TVT.2020.2986005).
- [48] D. Morimoto, Y. Iwamoto, M. Hiraga, and K. Ohkura, "Generating collective behavior of a multi-legged robotic swarm using deep reinforcement learning," *J. Robot. Mechatronics*, vol. 35, no. 4, pp. 977–987, Aug. 2023, doi: [10.20965/jrm.2023.p0977](https://doi.org/10.20965/jrm.2023.p0977).
- [49] Y. Cao, K. Ni, T. Kawaguchi, and S. Hashimoto, "Path following for autonomous mobile robots with deep reinforcement learning," *Sensors*, vol. 24, no. 2, p. 561, Jan. 2024, doi: [10.3390/s24020561](https://doi.org/10.3390/s24020561).
- [50] X. Zhu, F. Zhang, and H. Li, "Swarm deep reinforcement learning for robotic manipulation," *Proc. Comput. Sci.*, vol. 198, pp. 472–479, Apr. 2022, doi: [10.1016/j.procs.2021.12.272](https://doi.org/10.1016/j.procs.2021.12.272).
- [51] B. Sangiovanni, A. Rendiniello, G. P. Incremona, A. Ferrara, and M. Piastra, "Deep reinforcement learning for collision avoidance of robotic manipulators," in *Proc. Eur. Control Conf. (ECC)*, Limassol, Cyprus, Jun. 2018, pp. 2063–2068, doi: [10.23919/ECC.2018.8550363](https://doi.org/10.23919/ECC.2018.8550363).
- [52] B. Pintos Gómez de las Heras, R. Martínez-Tomás, and J. M. Cuadra Troncoso, "Self-learning robot autonomous navigation with deep reinforcement learning techniques," *Appl. Sci.*, vol. 14, no. 1, p. 366, Dec. 2023, doi: [10.3390/app14010366](https://doi.org/10.3390/app14010366).
- [53] S. S. Samsani, H. Mutahira, and M. S. Muhammad, "Memory-based crowd-aware robot navigation using deep reinforcement learning," *Complex Intell. Syst.*, vol. 9, no. 2, pp. 2147–2158, Nov. 2022, doi: [10.1007/s40747-022-00906-3](https://doi.org/10.1007/s40747-022-00906-3).
- [54] L. Chen, Z. Jiang, L. Cheng, A. C. Knoll, and M. Zhou, "Deep reinforcement learning based trajectory planning under uncertain constraints," *Frontiers Neurobotics*, vol. 16, May 2022, Art. no. 883562, doi: [10.3389/fnbot.2022.883562](https://doi.org/10.3389/fnbot.2022.883562).
- [55] E. Solowjow and J. Aparicio, "Reinforcement learning shows promise for industrial robotics," *Robot Rep.*, Dec. 2018. [Online]. Available: <https://www.therobotreport.com/reinforcement-learning-industrial-robotics/>
- [56] M. A. Akhloufi, S. Arola, and A. Bonnet, "Drones chasing drones: Reinforcement learning and deep search area proposal," *Drones*, vol. 3, no. 3, p. 58, Jul. 2019, doi: [10.3390/drones3030058](https://doi.org/10.3390/drones3030058).
- [57] A. A. Darwish and A. Nakhmani, "Drone navigation and target interception using deep reinforcement learning: A cascade reward approach," *IEEE J. Indoor Seamless Positioning Navigat.*, vol. 1, pp. 130–140, 2023, doi: [10.1109/JISPIN.2023.3334690](https://doi.org/10.1109/JISPIN.2023.3334690).
- [58] M. S. Assenine, W. Bechkit, I. Mokhtari, H. Rivano, and K. Benatchba, "Cooperative deep reinforcement learning for dynamic pollution plume monitoring using a drone fleet," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 7325–7338, Mar. 2024, doi: [10.1109/JIOT.2023.3328242](https://doi.org/10.1109/JIOT.2023.3328242).
- [59] I. M. A. Nahrendra, C. Tirtawardhana, B. Yu, E. M. Lee, and H. Myung, "Retro-RL: Reinforcing nominal controller with deep reinforcement learning for tilting-rotor drones," *IEEE Robot. Autom. Lett.*, vol. 7, no. 4, pp. 9004–9011, Oct. 2022, doi: [10.1109/LRA.2022.3189446](https://doi.org/10.1109/LRA.2022.3189446).
- [60] K. Li, W. Ni, E. Tovar, and M. Guizani, "Deep reinforcement learning for real-time trajectory planning in UAV networks," in *Proc. IEEE Int. Wireless Commun. Mobile Comput. (IWCMC)*, 2020, pp. 958–963, doi: [10.1109/IWCMC48107.2020.9148316](https://doi.org/10.1109/IWCMC48107.2020.9148316).
- [61] J. Tang, Y. Liang, and K. Li, "Dynamic scene path planning of UAVs based on deep reinforcement learning," *Drones*, vol. 8, no. 2, p. 60, Feb. 2024, doi: [10.3390/drones8020060](https://doi.org/10.3390/drones8020060).
- [62] C. Yan, X. Xiang, and C. Wang, "Towards real-time path planning through deep reinforcement learning for a UAV in dynamic environments," *J. Intell. Robotic Syst.*, vol. 98, no. 2, pp. 297–309, Sep. 2019, doi: [10.1007/s10846-019-01073-3](https://doi.org/10.1007/s10846-019-01073-3).
- [63] X. Kong, Y. Zhou, Z. Li, and S. Wang, "Multi-UAV simultaneous target assignment and path planning based on deep reinforcement learning in dynamic multiple obstacles environments," *Frontiers Neurobotics*, vol. 17, Jan. 2024, Art. no. 1302898, doi: [10.3389/fnbot.2023.1302898](https://doi.org/10.3389/fnbot.2023.1302898).
- [64] S. Ouahouh, M. Bagaa, J. Prados-Garzon, and T. Taleb, "Deep-reinforcement-learning-based collision avoidance in UAV environment," *IEEE Internet Things J.*, vol. 9, no. 6, pp. 4015–4030, Mar. 2022, doi: [10.1109/JIOT.2021.3118949](https://doi.org/10.1109/JIOT.2021.3118949).
- [65] S. Song, Y. Zhang, X. Qin, K. Saunders, and J. Liu, "Vision-guided collision avoidance through deep reinforcement learning," in *Proc. IEEE Nat. Aerosp. Electron. Conf.*, Dayton, OH, USA, Aug. 2021, pp. 191–194, doi: [10.1109/NAECON49338.2021.9696380](https://doi.org/10.1109/NAECON49338.2021.9696380).
- [66] S.-Y. Shin, Y.-W. Kang, and Y.-G. Kim, "Obstacle avoidance drone by deep reinforcement learning and its racing with human pilot," *Appl. Sci.*, vol. 9, no. 24, p. 5571, Dec. 2019, doi: [10.3390/app9245571](https://doi.org/10.3390/app9245571).
- [67] S. Anbalagan, P. Srividya, B. Thilaksurya, S. G. Senthivel, G. Suganeshwari, and G. Raja, "Vision-based ingenious lane departure warning system for autonomous vehicles," *Sustainability*, vol. 15, no. 4, p. 3535, Feb. 2023, doi: [10.3390/su15043535](https://doi.org/10.3390/su15043535).
- [68] A. Ahadi-Sarkani and S. Elmalaki, "ADAS-RL: Adaptive vector scaling reinforcement learning for human-in-the-loop lane departure warning," in *Proc. 1st Int. Workshop Cyber-Phys.-Hum. Syst. Design Implement.*, May 2021, pp. 13–18, doi: [10.1145/3458648.3460008](https://doi.org/10.1145/3458648.3460008).
- [69] B. Brik and A. Ksentini, "Toward optimal MEC resource dimensioning for a vehicle collision avoidance system: A deep learning approach," *IEEE Netw.*, vol. 35, no. 3, pp. 74–80, May 2021, doi: [10.1109/MNET.011.2000577](https://doi.org/10.1109/MNET.011.2000577).
- [70] M. S. Farooq et al., "A conceptual multi-layer framework for the detection of nighttime pedestrian in autonomous vehicles using deep reinforcement learning," *Entropy*, vol. 25, no. 1, p. 135, Jan. 2023, doi: [10.3390/e25010135](https://doi.org/10.3390/e25010135).
- [71] D. C. Selvaraj, S. Hegde, N. Amati, F. Deflorio, and C. F. Chiasserini, "A deep reinforcement learning approach for efficient, safe and comfortable driving," *Appl. Sci.*, vol. 13, no. 9, p. 5272, Apr. 2023, doi: [10.3390/app13095272](https://doi.org/10.3390/app13095272).

- [72] D. C. Selvaraj, S. Hegde, N. Amati, F. Deflorio, and C. F. Chiasserini, "An ML-aided reinforcement learning approach for challenging vehicle maneuvers," *IEEE Trans. Intell. Vehicles*, vol. 8, no. 2, pp. 1686–1698, Feb. 2023, doi: [10.1109/TIV.2022.3224656](https://doi.org/10.1109/TIV.2022.3224656).
- [73] Y. Lin, J. McPhee, and N. L. Azad, "Comparison of deep reinforcement learning and model predictive control for adaptive cruise control," *IEEE Trans. Intell. Vehicles*, vol. 6, no. 2, pp. 221–231, Jun. 2021, doi: [10.1109/TIV.2020.3012947](https://doi.org/10.1109/TIV.2020.3012947).
- [74] K. V. Sakhare, T. Tewari, and V. Vyas, "Review of vehicle detection systems in advanced driver assistant systems," *Arch. Comput. Methods Eng.*, vol. 27, no. 2, pp. 591–610, Mar. 2019, doi: [10.1007/s11831-019-09321-3](https://doi.org/10.1007/s11831-019-09321-3).
- [75] *AI & Robotics*. Tesla. Accessed: Jan. 10, 2025. [Online]. Available: <https://www.tesla.com/AI>
- [76] Y. Qian, S. Feng, W. Hu, and W. Wang, "Obstacle avoidance planning of autonomous vehicles using deep reinforcement learning," *Adv. Mech. Eng.*, vol. 14, no. 12, Dec. 2022, Art. no. 168781322211396, doi: [10.1177/16878132221139661](https://doi.org/10.1177/16878132221139661).
- [77] A. Jatavallabha, "Tesla's autopilot: Ethics and tragedy," 2024, *arXiv:2409.17380*.
- [78] M. Bansal, A. Krizhevsky, and A. Ogale, "ChauffeurNet: Learning to drive by imitating the best and synthesizing the worst," 2018, *arXiv:1812.03079*.
- [79] (Dec. 2024). *Are Waymo's Autonomous Taxis Safer Than Human-driven Vehicles? New Study Says So*. Designboom—architecture design magazine. [Online]. Available: <https://www.designboom.com/technology/waymo-autonomous-taxis-safer-than-human-driven-vehicles-study-swiss-re-12-23-2024/>
- [80] *Safety-Waymo*. Waymo. Accessed: Jan. 12, 2025. [Online]. Waymo. Available: <https://waymo.com/safety/>
- [81] B. Heater. (Apr. 2024). *Boston Dynamics' Atlas Humanoid Robot Goes Electric*. TechCrunch. [Online]. Available: <https://techcrunch.com/2024/04/17/boston-dynamics-atlas-humanoid-robot-goes-electric/>
- [82] J. Del Rey. (Dec. 2019). *How Robots are Transforming Amazon Warehouse Jobs for Better and Worse*. Vox. [Online]. Available: <https://www.vox.com/recode/2019/12/11/20982652/robots-amazon-warehouse-jobs-automation>
- [83] N. Savushkin. (2024). *Warehouse Automation in Logistics?: Case Study of Amazon and Ocado*. Theseus. [Online]. Available: <https://urn.fi/URN:NBN:fi:amk-2024052114014>
- [84] A. Staff. (Jun. 2022). *Amazon Prime Air Prepares for Drone Deliveries*. [Online]. Available: <https://www.aboutamazon.com/news/transportation/amazon-prime-air-prepares-for-drone-deliveries>
- [85] (2022). *Package Delivery By Drone (Part 135)*. Federal aviation administration. [Online]. Available: https://www.faa.gov/uas/advanced_operations/package_delivery_drone
- [86] W. Hlophe. (Dec. 2023). *Flight of the Future: Exploring DJI's AI-Driven Drone Evolution - Digital Innovation and Transformation*. Digital innovation and transformation. [Online]. Available: <https://d3.harvard.edu/platform-digit/submission/flight-of-the-future-exploring-djis-ai-driven-drone-evolution/>
- [87] (Aug. 2024). *DJI Drone With Collision Avoidance-Neuvition | Collision Avoidance Systems*. [Online]. Available: <https://www.neuvition.com/dji-drone-with-collision-avoidance-neuvition/>
- [88] *Mobileye ADAS Platforms | Vision-centric Intelligent Safety on the Road*. Mobileye. Accessed: Jan. 15, 2025. [Online]. Available: <https://www.mobileye.com/solutions/adas/>
- [89] A. Shashua. (May 2020). *The Challenge of Supporting AV at Scale | Mobileye Opinion*. Mobileye. [Online]. Available: <https://www.mobileye.com/opinion/the-challenge-of-supporting-av-at-scale/>
- [90] *Mobileye Improves Deep Learning Training Performance and Reduces Costs Using Amazon EC2 DL1 Instances—Mobileye Case Study—AWS*. Amazon Web Services. Accessed: Jan. 16, 2025. [Online]. Available: <https://aws.amazon.com/solutions/case-studies/mobileye-ec2-dl1-case-study/>
- [91] (May 2019). *NVIDIA DRIVE Platform for Self-Driving Overview—AutoPilot Review*. [Online]. Available: <https://www.autopilotreview.com/nvidia-drive-platform/>



Shruti Govinda received the bachelor's degree in mathematics from the University of Sharjah in 2023, where she is currently pursuing the master's degree in data science. As a Research Assistant at the University of Sharjah, she contributed to mathematical analysis in complex systems. Her master's research includes projects on Dubai house price prediction using machine learning, sentiment analysis of ChatGPT early adopters, and ridge-type estimation for conditional autoregressive (CAR) models. She is also exploring deep learning techniques for autonomous systems. Her research interests include deep learning applications in autonomous systems, machine learning for spatial modeling, and artificial intelligence in scientific computing. She has research experience at Sharjah Academy for Astronomy, Space Sciences and Technology (SAASST), where she worked on deep learning for meteorite recognition and meteor detection.



Bouziane Briki (Senior Member, IEEE) received the Engineering degree (Hons.) in computer science and the M.Sc. and Ph.D. degrees from Laghouat University, Algeria, in 2010, 2013, and 2017, respectively. He is currently an Assistant Professor with the Computer Science Department, Computing and Informatics College, Sharjah University, United Arab Emirates. Before joining Sharjah University, he was an Assistant Professor with the DRIVE Department, Bourgogne University, France. He was also a Post-Doctoral with the CESI School, University of Troyes, and Eurecom Research Institute, France. He has been working on resources management and security challenges of 5G network slicing in the context of H2020 European projects, including MonB5G, 5GDrones, InDiD, and 5G-INSIGHT. His research interests include 5G and beyond networks, explainable AI, and machine/deep learning for wireless networks.



Saad Harous (Senior Member, IEEE) received the Ph.D. degree in computer science from Case Western Reserve University, Cleveland, OH, USA, in 1991. He has more than 35 years of experience in teaching and research in three different countries: USA, Oman, and United Arab Emirates. He is currently a Professor at the College of Computing and Informatics, University of Sharjah, United Arab Emirates. He has published more than 250 journals and conference papers. His teaching interests include programming, data structures, design and analysis of algorithms, operating systems, and networks. His research interests include parallel and distributed computing, P2P delivery architectures, wireless networks, and the use of computers in education and processing Arabic language.